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Key Points:
- An in-depth review of radar-rain gauge merging methods is presented, which compiles much of the work undertaken in this area to date
- It includes a categorization of methods, summary of merging intercomparison studies to date, and factors affecting merging performance
- The potential and challenges of applying merging at the fine spatial-temporal resolutions required for urban hydrology are discussed

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A Review of Radar-Rain Gauge Data Merging Methods and Their Potential for Urban Hydrological Applications

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Abstract Radar-rain gauge merging techniques have been widely used to improve the applicability of radar and rain gauge rainfall estimates by combining their advantages, while partially overcoming their individual weaknesses. Despite significant research in this area, guidance on the suitability of and factors affecting merging techniques at the fine spatial-temporal resolutions required for urban hydrological applications is still insufficient. In this paper, an in-depth review of radar-rain gauge merging techniques is conducted, with a focus on their potential for urban hydrological applications. An overview is first given of existing merging techniques and an application-oriented categorization is proposed: (1) radar bias adjustment methods, (2) rain gauge interpolation methods using radar spatial association as additional information, and (3) radar-rain gauge integration methods. A detailed review is given of studies focusing on the evaluation and intercomparison of merging methods, based upon which the most widely used and best performing techniques from each category are identified. These are mean field bias adjustment, kriging with external drift, and Bayesian merging. Climatological, operational, and methodological factors affecting merging performance are then reviewed and their relevance for urban applications discussed. Based on this review, conclusions on merging potential for urban applications are drawn and research gaps are identified, which should be addressed to provide further guidance on the use of merging techniques for urban hydrological applications.

1. Introduction

The two sensors commonly used for rainfall estimation at catchment scales are rain gauges and weather radars (Cole & Moore, 2008). Rain gauges provide relatively accurate point rainfall estimates near the ground surface; nonetheless, they cannot well capture the spatial variability of rainfall, which has a significant impact on hydrological systems and thus on runoff modeling, particularly in the case of small urban catchments (Del Giudice et al., 2013; Gires et al., 2012; Ochoa-Rodriguez et al., 2015; Schellart et al., 2012). Moreover, since dense rain gauge networks cannot cover large areas, it is difficult to forecast rainfall with longer lead time based on rain gauge data only (Looper & Vieux, 2012; Simões et al., 2011; Wang et al., 2011). In contrast, radars can survey large areas and can better capture the spatial variability of rainfall fields, thus improving the short-term predictability of rainfall (i.e., rainfall nowcasting a few hours ahead, generally up to 6 hr). However, the accuracy of radar measurements is in general insufficient, particularly in the case of extreme rainfall magnitudes (Bárdossy & Pegram, 2017; Einfalt et al., 2004; Einfalt et al., 2005; Marra & Morin, 2015). This is due to the fact that instead of being a direct measurement, radar rainfall intensity is derived indirectly from measured radar reflectivity. As a result, both radar reflectivity measurements and the reflectivity-intensity conversion process are subject to multiple sources of error (Harrison et al., 2009; Krajewski et al., 2010; Villarini & Krajewski, 2010). A number of corrections can be applied in order to reduce these errors, and great progress has been made in this direction in the last few decades (see review of progress to date in Krajewski et al., 2010, and Thorndahl et al., 2017), including more recent advances based upon dual-polarization radar measurements (Bringi et al., 2011; Bringi & Chandrasekar, 2001; Chandrasekar et al., 2013; Hall et al., 2015; Rico-Ramirez & Cluckie, 2008; Sugier & Tabary, 2006). However, it is virtually impossible to have error-free radar quantitative precipitation estimates (QPEs) and the new radar technologies and processing techniques will not change the inherent limitations of radar as a rainfall measurement tool, such as the fact that rainfall is measured indirectly, often well above the ground and often far away from the radar, which results in beam broadening and range degradation. The

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insufficient accuracy of radar QPEs has proven problematic and has hindered their widespread practical use (Berne & Krajewski, 2013; McKee & Binns, 2015; Schellart et al., 2012).

To improve the accuracy of radar QPEs while preserving their spatial description of rainfall fields, it is possible to adjust them based on rain gauge measurements; doing so enables combining the advantages of each sensor, while partially overcoming their drawbacks. Gauge-based adjustment of radar QPEs, also referred to as radar-rain gauge combination or merging, has been an active topic of research ever since the potential of radars as precipitation measuring devices was realized in the 1940s and 1950s (e.g., Brandes, 1975; Hitschfeld & Bordan, 1954; Wilson, 1970). Several radar-rain gauge merging techniques have been developed, which have proven effective to improve the accuracy of radar QPEs and hence their applicability for hydrological uses (Goudenhoofdt & Delobbe, 2009; Looper & Vieux, 2012; McKee & Binns, 2015; Smith et al., 2007; Vieux & Bedient, 2004; Wang et al., 2013). While significant progress has been made in this direction, a number of questions remain, including the full potential of applying radar-rain gauge merging at the spatial-temporal resolutions required for urban hydrology.

The relatively small size of urban catchments (as compared to river basins) alongside their high degree of imperviousness and associated fast response to rainfall make them very sensitive to the spatial and temporal variability of precipitation (Aronica & Cannarozzo, 2000; Segond et al., 2007; Syed et al., 2003; Villarini et al., 2010). As a result, urban hydrological applications generally require high-quality rainfall estimates, in terms of both resolution and accuracy. Among urban hydrological applications, detailed urban drainage modeling, including simulation of runoff flows in the overland and/or underground drainage systems, has the most stringent rainfall data requirements (Berne et al., 2004; Einfalt, 2005; Einfalt et al., 2004; Fabry et al., 1994; Schellart et al., 2012; Schilling, 1991; see Figure 1). A recent multistorm, multicatchment investigation by Ochoa-Rodriguez, Wang, Gires, et al. (2015) into rainfall resolution requirements for urban drainage modeling identified critical rainfall spatial-temporal resolutions of the order of 1 km and 1-5 min. What is more, the use of detailed urban drainage models and hence the need for high resolution and accuracy rainfall data has significantly increased in recent years and is expected to continue to increase, fueled by developments in Geographic Information Systems, ever increasing data availability and computational power (Bailey & Margetts, 2008; Fewtrell et al., 2011; Giangola-Murzyn et al., 2012; Hunter et al., 2008; Pina et al., 2014; Salvadore et al., 2015). This has, in turn, led to increasingly common use of radar rainfall estimates in urban hydrology, as a complement to traditional rain gauge networks (Thorndahl et al., 2017). Consequently, radar-rain gauge combination has become more relevant for urban applications.

This paper focuses on the more stringent urban hydrological applications, such as detailed urban drainage modeling, which, as mentioned above, require rainfall data at resolutions of the order of 1-5 min in time and 1 km or finer in space. From here on these scales and resolutions will be referred to as the "urban scales and resolutions of interest." These scales and resolutions are significantly finer than those at which radar-rain gauge merging has traditionally been applied, with typical merging domains of interest covering entire regions, if not countries, and spatial temporal resolutions generally of the order of, and often coarser than 1 km^2 in space and 1 hr in time (Wang et al., 2013).

The application of radar-rain gauge merging at the fine spatial-temporal scales required for urban applications poses specific challenges. These include (i) the fact that spatial-temporal sampling differences between radar and rain gauge measurements are magnified at fine scales, (ii) the need to preserve small-scale (e.g., convective) features in the merged estimates, (iii) the often limited availability of rain gauges across urban areas, (iv) and the increasing computational requirements associated with the application of merging at higher spatial-temporal resolution, among others. As such, when looking at applying merging techniques at urban scales, it is important to consider these factors, to understand the requirements of the different merging methods that are available and their ability to handle the challenges associated with small-scale applications.

With the purpose of exploring the potential of merging techniques at the fine spatial-temporal scales required for urban hydrological applications, in this paper a detailed review of merging techniques is presented and the implications for the most stringent urban applications (as defined above) are discussed. The review compiles much of the work undertaken in the area of radar-rain gauge merging to date and will assist in the selection of merging methods and methodological choices for urban applications. Additionally, it will help identify research gaps, which should be addressed to provide further guidance on the use of merging techniques for urban hydrological applications.

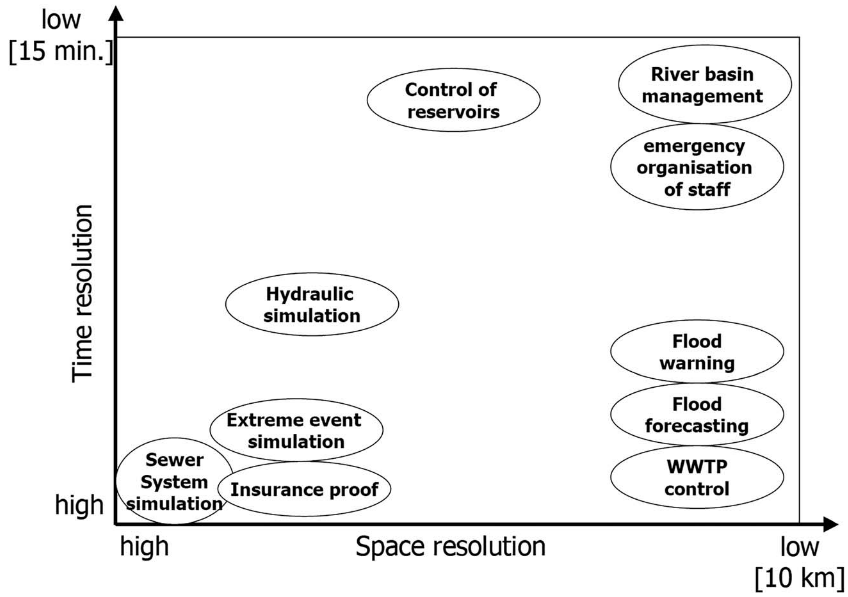

Figure 1. Rainfall data resolution requirements for different urban hydrological applications (Einfalt, 2005).

The paper starts with an overview of radar and rain gauge rainfall estimates and the uncertainties associated with them, which ultimately condition merging performance and play an important role in the selection of merging methods for a given application. This is followed by the actual review, which includes the following:

1. A review and categorization of existing merging techniques.
2. A review of studies focusing on the evaluation and intercomparison of merging methods, including studies at large and small scales. Based upon this, the best performing and most widely used techniques are identified, and their features and relative performance are further discussed in light of results to date.
3. A review of climatological and operational factors, which impact merging performance.
4. A review of methodological choices and approaches, which not only impact merging performance but which may also help deal with some of the challenges associated with the application of merging at fine spatial-temporal scales.

The review is followed by a discussion of the potential for application of merging techniques at urban scales, considering how different techniques, methodological choices, and climatological and operational factors may impact merging performance at the spatial-temporal scales of interest. The paper concludes with a summary of the main findings from the review and identified research gaps, which should be addressed to provide further guidance on the use of merging techniques for urban hydrological applications.

2. Radar and Rain Gauge Rainfall Estimates: An Overview

With the purpose of providing some background to the present review, an overview of rain gauge and radar rainfall estimates is given next, including a description of the nature of the estimates provided by each sensor and the uncertainties associated with them. The main differences between radar and rain gauge rainfall estimates, which should be kept in mind when looking at merging data from the two sensors, are also discussed in this section.

2.1. Rain Gauge Rainfall Estimates

“Rain gauge” is a broad term, which can be used to refer to any instrument employed to measure the amount of liquid precipitation over a set period. In the context of this paper, the term rain gauge will be generally used to refer to instruments used to measure the amount of liquid precipitation at a point location, over a set period of time, mainly by means of direct rainfall collection (i.e., catching rain gauges; Lanza et al., 2005). The potential use of data from noncatching point rain gauges (e.g., from optical, acoustic, and piezoelectric disdrometers), including from low-cost sensors utilized, for example, in citizen observatories, is discussed in the conclusions and outlook section of the paper. Below, rain gauge will therefore refer to "catching rain gauge."

Rain gauges are the most widely used rainfall sensors, providing low-cost, direct, and relatively accurate point rainfall estimates near the ground surface. In fact, rain gauge records are often used as ground truth in the calibration and verification of remote rainfall sensors such as weather radars (Habib et al., 2001; Upton & Rahimi, 2003). Rain gauges can be further categorized into nonrecording and recording gauges. The former collect rainfall volumes over a given period, which are then measured manually, while the latter provide automatic records of rainfall by means of mechanical and/or electronic mechanisms and can provide rainfall estimates at finer temporal resolution as compared to nonrecording gauges. Examples of recording catching gauges are the tipping bucket, weighing, and level gauges. Results of an experimental intercomparison of recording gauges can be found in Lanza et al. (2005). Recording gauges are generally the preferred choice for rainfall monitoring at catchment scale, while nonrecording gauges are often used as reference for quality control of recording rain gauges (EA, 2013). Tipping bucket rain gauges (TBRs) are the most common type of automatic recording gauge and are in fact the main data source used for adjustment of radar rainfall estimates (e.g., Borup et al., 2016; Jewell & Gaussiat, 2015; Nanding et al., 2015; Thorndahl et al., 2014; Wang et al., 2013). Weighing gauges are however becoming increasingly popular, owing mainly to their lower maintenance requirements as compared to TBRs (Instrumentation Monthly, 2013), while displaying a performance comparable to that of TBRs (Colli et al., 2014). For example, national rainfall monitoring networks in Canada and the UK now include a significant proportion of weighing gauges (Instrumentation Monthly, 2013; McKee & Binns, 2015). When using catching rain gauge records, it is important to keep in mind the nature of their measurements; that is, rainfall accumulations at point locations during the time interval between measurements—for example, between tips in the case of TBRs or between weight records in the case of weighing gauges, with bucket resolution and polling frequency, respectively, influencing the effective temporal resolution and accuracy of TBR and weighing rain gauge estimates (Colli et al., 2014).

Uncertainties in Rain Gauge Rainfall Estimates

Rain gauge measurements are subject to two main sources of error: (1) instrumental errors, which include systematic, local random, and malfunction errors, and (2) spatial sampling errors (SSEs), related to the spatial representativeness of point rain gauge measurements. Accounting for and, if possible, reducing the uncertainties in rain gauge measurements is critical in radar-rain gauge merging, particularly as rain gauge measurements are typically used as approximations of the true rainfall and, as a result, as reference for valuation and adjustment of radar QPEs.

While rain gauge records are usually regarded as reliable, they may contain significant instrumental errors arising from a variety of sources. In order of general importance, systematic errors common to all rain gauges include errors due to wind effects (including wind deformation due to poor siting and localized field deformation above the gauge orifice), wetting loss in the internal walls of the collector, evaporation from the container, and errors due to in and out-splashing of water (Lanza et al., 2005; WMO, 2008). In addition, TBRs are known to systematically underestimate rainfall at higher intensities because of the rainwater amount that is lost during the tipping movement of the bucket (La Barbera et al., 2002). This is a systematic error unique to the TBR and is of the same order of importance as wind-induced losses. The order of magnitude of the two main systematic error sources associated with TBRs is 2-10% for wind-induced losses (depending on wind speed, weight of precipitation, gauge construction parameters, and location; Sevruk & Nespor, 1998) and up to 10% for rainfall intensities of 100 mm/hr and 20% for intensities of 200 mm/hr, for water losses during the tipping action (Luyckx & Berlamont, 2001; Molini et al., 2005). Errors due to water loss during the tipping action can be largely corrected based upon results of dynamic calibration (Luyckx & Berlamont, 2001; Sene, 2010). Besides systematic errors, TBR records are also subject to local random errors, mainly related to their discrete sampling mechanism (Ciach, 2003; Habib et al., 2001; Habib et al., 2008); such errors are influenced by bucket resolution, rainfall rate, and intermittency and can be of the order of 6% for rain rates of around 10 mm/hr at the 5-min time scale (Ciach, 2003; Colli et al., 2014). Weighing gauges have also been found to be subject to similar local random errors, related to the response time of the measurement system (Colli et al., 2014). Additional errors in rain gauge measurements may arise from operational conditions such as miscalibration, blockages, and double-tipping (in the case of TBRs). Operational errors can usually be detected (and therefore removed or corrected) through time series analysis or through comparison against

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neighboring gauges and even against co-located radar estimates (Anagnostou et al., 1998; Upton & Rahimi, 2003). Given that rain gauge records are often used as ground truth, it is important to ensure that they have been quality-controlled before they are used in radar-rain gauge merging or in any other hydrological application; doing so can largely improve subsequent results (Looper & Vieux, 2012).

The SSE is defined as the error resulting from approximating an areal estimate using point measurements (Villarini et al., 2008). With the typical catch area of a rain gauge being of the order of 20 cm and given the high spatial variability of rainfall fields, SSE is evident both when a single rain gauge is used to approximate the areal rainfall across an area as small as a 1-km radar pixel (Gires et al., 2014; Jensen & Pedersen, 2005; Wood et al., 2000a), as well as when single or multiple rain gauges are used to approximate the areal rainfall field across larger catchment areas (Rodríguez-Iturbe & Mejía, 1974; Seed & Austin, 1990;Villarini, Mandapaka, et al., 2008; Wood et al., 2000a). Generally, SSE has been found to decrease substantially with increasing aggregation time (Villarini, Mandapaka, et al., 2008; Wood et al., 2000a) adding to the complications associated with fine resolution rainfall estimation and merging. In addition to the aggregation time, other factors impacting SSE include gauge density and distribution, rainfall patterns, and orography (Huff, 1970; Langousis & Kaleris, 2013; Villarini, Mandapaka, et al., 2008; Wood et al., 2000a). For example, on the topic of temporal aggregation and rain gauge density impact on SSE, based on records from a dense test network of rain gauges, Villarini, Mandapaka, et al. (2008) concluded that to estimate areal rainfall across a 200 km^2 area within 20% of its true value at temporal aggregation scales of 15 min, 1 hr, 3 hr, and 1 day, the number of rain gauges required would respectively be over 25, around 25, 15, and 4.

As can be expected, SSE can be particularly critical when rain gauge records are being compared against radar QPEs, for example, in the evaluation of radar QPEs using rain gauge estimates as reference or as is done throughout the radar-rain gauge merging process, with different merging methods entailing different ways of comparing radar and rain gauge measurements. In fact, the point-area spatial mismatch between rain gauges and radar measurements has been found to be a significant part of the overall differences between radar and rain gauge records, particularly at short time scales (Ciach & Krajewski, 1999a, 1999b; Gebremichael & Krajewski, 2004; Gires et al., 2014; Kitchen & Blackall, 1992; Seo & Krajewski, 2011; Zawadzki, 1975). A formal approach for partitioning the variance of the radar-rain gauge differences into the radar QPE estimation error variance and the rain gauge SSE variance (so-called rain gauge error variance) was introduced by Ciach and Krajewski (1999a). By applying this method to a quality-controlled data set consisting of five rain gauges and radar QPEs at 4-×4-km^2 resolution across a 12-×12-km^2 domain, Ciach and Krajewski (1999a) estimated rain gauge error variance across multiple temporal aggregation scales. For accumulation times of 5 min and 1 hr, rain gauge error variance was estimated to make up approximately 70% and 50%, respectively, of the overall radar-rain gauge discrepancies. Results of a similar order of magnitude were later reported by Habib and Krajewski (2002), who applied Ciach and Krajewski's error variance decomposition approach to a rich radar-rain gauge data set across Florida, USA. Other methods for estimating and/or dealing with rain gauge SSE at the radar pixel scale, to aid the comparison of radar and rain gauge measurements, include (1) quantification of subpixel spatial variability, for example, making use of spatial downscaling techniques (Gires et al., 2014), and (2) synthetization of gridded rainfall fields based on point rain gauge measurements prior to the radar-rain gauge comparison, for example, through block-kriging interpolation (Chiles & Delfiner, 2012; Todini, 2001).

2.2. Radar Rainfall Estimates

Weather radars measure rainfall indirectly by sending pulses of electromagnetic radiation into the atmosphere and listening for return signals backscattered by rain droplets or other precipitation particles, with the quantity of reflected energy being a function of particle size, type and distribution. The received reflectivity can then be used to estimate rainfall rates at a given distance from the radar. Unlike rain gauges, radars provide instantaneous ("snapshot-like") volumetric measurements of rainfall (Figure 2). The effective spatial-temporal resolution and accuracy of radar QPEs are determined by radar hardware, scanning strategy, distance from the radar, atmospheric conditions, and correction and rainfall estimation algorithms. An overview of these factors and of uncertainties in radar rainfall estimates is given next. For additional information, the reader may refer to weather radar textbooks and publications focusing on radar rainfall estimation (Berne & Krajewski, 2013; Bringi & Chandrasekar, 2001; Hong & Gourley, 2015; Krajewski & Smith, 2002; Villarini & Krajewski, 2010; WMO, 2009).

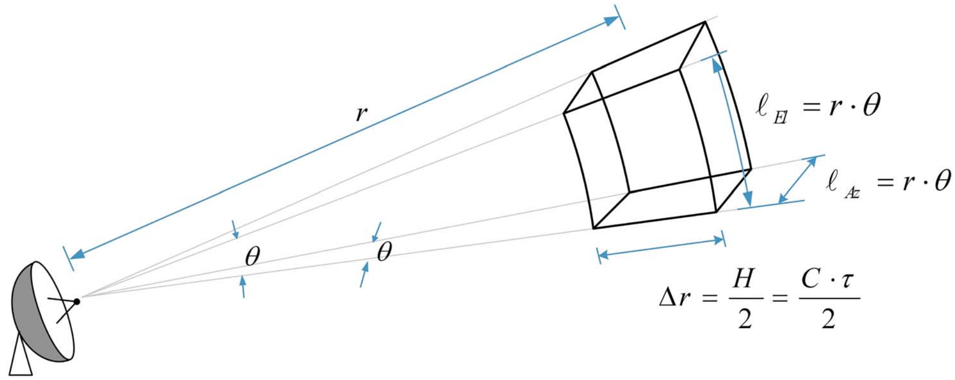

Figure 2. Radar scanning volume.

2.2.1. Radar Hardware and Scanning Strategy

For a typical weather radar, a parabolic antenna focuses pulsed, polarized electromagnetic radiation of a given frequency (f) and wavelength (λ) into a narrow pencil beam of angular width θ. Each radiation pulse (of duration τ and length H), and travelling at the speed of light C), is backscattered by the precipitation it interacts with. A radar completes a scan at regular time intervals and, depending on the scanning strategy and hardware, scans may be carried out at different elevations (Hong & Gourley, 2015; WMO, 2009).

According to the frequency band —and hence the wavelength— at which they operate, weather radars can be of three main types (from higher to lower frequency): X-band, C-band, and S-band. The higher the frequency, the shorter the wavelength ($\lambda = C/f$), which means that higher frequency radars can detect smaller particles. However, high-frequency signals are also associated with lower power, which results in shorter radar range and higher susceptibility to attenuation by strong rainfall intensities. On the other hand, higher frequency signals require smaller antenna dishes in order to focus the radiation into a narrow pencil beam. Therefore, high-frequency radars are smaller and cheaper, but also less powerful, than low-frequency ones. Low-frequency S-band radars are often used to monitor severe weather phenomena in long ranges, such as tropical storms. These radars are widely used in the United States (Chen & Chandrasekar, 2015; Crum et al., 1993; Fulton et al., 1998). Intermediate frequency C-band radars provide a good compromise between cost and power and are the most widely used in Europe (Huuskonen et al., 2014). High-frequency X-band radars are the smallest (often portable) and cheapest of all and are often used to study cloud development (as they are sensitive to small rain particles) and to monitor rainfall over small areas of interest, including in local urban areas not covered or well served by other (bigger) radars and in between narrow valleys. In fact, the use of X-band radars for urban rainfall monitoring has been an active topic of research in recent years (Allegretti et al., 2012; Berenguer et al., 2012; Chandrasekar et al., 2009; Li & Willems, 2017; Lo Conti et al., 2015; Ochoa-Rodríguez et al., 2014; Paz et al., 2018; ten Veldhuis et al., 2014; van de Beek et al., 2010; Yoon & Nakakita, 2017).

The modulated length (H) of the radiation pulses emitted by the radar is chosen by radar operators and must represent a balance between spatial resolution, sensitivity, and desired range coverage. Shorter pulses yield higher resolution data in the range (radial) coordinate (see Figure 2), but the power returned once they hit a target is also lower and it may be fully extinct over long ranges, thus limiting effective radar coverage. Pulse lengths typically range between ~10 m and 1 km. European radars usually operate at pulse durations of 2 μs, which is equivalent to ~600 m, although the use of shorter pulse (0.5 μs–150 m) scanning strategies has been tested with the aim of providing higher spatial resolution radar QPEs based on existing C-band radar networks (Ochoa-Rodriguez et al., 2015a). U.S. radars usually operate at larger pulses (between 1.75-4.7 μs and 470-1410 m), so as to ensure larger coverage.

The polarization of radar signal refers to the orientation of the magnetic field. At present, most weather radars transmit and receive radio waves with a single horizontal polarization. Such radars are referred to as single-polarization radars. In contrast, dual-polarization or polarimetric radars, which are becoming increasingly popular, can transmit and receive both horizontal and vertical polarized signals. By emitting signals with two polarizations, polarimetric radars can obtain additional information (e.g., differential reflectivity and differential phase), which enables a better characterization of hydrometeors and can improve the accuracy of the radar rainfall estimates (Bringi & Chandrasekar, 2001; Chandrasekar et al., 2013).

In addition to the above features, some radars are equipped with Doppler functionality, which enables them to not only detect the distance of the "target" but also its velocity toward or away from the radar. Doppler velocities can be used to better differentiate targets and to detect severe weather, such as rotation within supercell thunderstorms (Mitchell et al., 1998). Depending on whether radars are equipped with this functionality, they are referred to as Doppler or non-Doppler radars. Polarimetric Doppler radars are the most sophisticated type of radars and their use is quickly expanding (Berne & Krajewski, 2013; Bringi & Chandrasekar, 2001; Huuskonen et al., 2014; Wang et al., 2016).

2.2.2. Spatial-Temporal Resolution of Radar QPEs

As indicated above, in terms of space, radars carry out volumetric scans of the atmosphere (Figure 2). The effective resolution of the scanned volume (V) is determined by the range and angular resolutions.

The range resolution (Δr) is the resolution in the radial or range direction from the radar. Given that radars can effectively distinguish targets separated by at least one-half the pulse length, the range resolution corresponds to half the pulse length: $\Delta r = H/2$ (WMO, 2009). The range resolution remains constant at varying distances from the radar.

The angular resolution corresponds to the arc lengths in the vertical (i.e., elevation; ℓ_{El}) and azimuthal (ℓ_{Az}) directions at a given distance from the radar; assuming that the radar beam is circular (which is usually the case), these two arc lengths are the same. The angular resolution is therefore determined by the beam width (θ; which is a function of antenna size and frequency, as described above) and by the distance from the radar (r). For a beam width of 1° (which is a common beam width), azimuthal angular resolutions at different distances from the radar are as follows: ~175 m at 10 km, ~436 m at 25 km, ~873 m at 50 km, ~1.75 km at 100 km, and ~2.62 km at 150 km.

Radar QPE maps in polar coordinates are generally interpolated into a Cartesian grid of a given spatial resolution (e.g., 1×1 km^2). When using radar QPEs in Cartesian coordinates it is important to keep in mind the native resolution at which the estimates were derived. It is often the case that the Cartesian spatial resolution at which radar QPEs are provided is finer than the fundamental cell resolution of radar measurements over the area of interest; one must not be tricked by this.

In summary, the spatial resolution of radar QPEs is fundamentally determined by the pulse length (which is chosen by radar operators as a compromise between resolution versus radar coverage), beam width (which is a function of antenna size and signal frequency), and distance from the radar. While the spatial resolution of radar QPEs will indeed be smaller for higher frequency and constant dish size, lower frequency radars generally use larger dishes in order to generate a narrow beam, while high-frequency radars tend to have small dishes. As such, the resulting role of radar frequency on the spatial resolution of QPEs is small. The effective spatial resolution of radar QPEs is therefore mostly a matter of operational strategy and distance from the radar.

It is commonly—and wrongly—believed that X-band radars de facto provide higher spatial resolution QPEs than lower frequency radars (C-band and S-band); as explained above, this is not necessarily the case. The reason why higher spatial resolution rainfall estimates are often available from local X-band radars is that they are usually operated at short pulse lengths (which results in finer radial resolution) and their operational range is shorter than that of lower frequency radars (hence maximum distances from the radar remain relatively short and, as a result, the angular resolution of QPEs remains relatively fine). However, for a given distance from the radar and as long as the beam is narrow and the pulse short enough, high-resolution QPEs can also be obtained from lower frequency radars.

The temporal resolution of radar QPEs, on the other hand, is dictated by the rotational frequency of the radar and the duration of the scanning cycle. Smaller radars, such as local X-band radars, usually have higher rotational frequencies and shorter scanning cycles, therefore yielding higher temporal resolution radar QPEs. Radar QPEs from local X-band radars are usually available at 1-min temporal resolution, whereas QPEs provided by national meteorological services (based upon networks of C-and and S-band radars) are usually available at temporal resolutions ranging from 5 to 15 min (Berenguer et al., 2012; Chandrasekar et al., 2009; Chen & Chandrasekar, 2015; Harrison et al., 2009; Huuskonen et al., 2014; Ochoa-Rodriguez, Wang, Gires, et al., 2015; ten Veldhuis et al., 2014).

2.2.3. Derivation of Radar QPEs

Based upon the received radiation backscattered by the rainfall droplets contained within a scanning volume, a reflectivity value (Z), theoretically linked to raindrop size distribution (DSD) within the scan volume, can be estimated. A detailed description of how Z is estimated can be found in many radar textbooks (e.g., Bringi & Chandrasekar, 2001; Hong & Gourley, 2015). Reflectivity (Z) is then converted to rainfall rate (R), usually by means of a power law of the form $Z = a \cdot R^b$ (Marshall & Palmer, 1948), where a and b are parameters, which can also be theoretically linked to rain DSD and are generally deduced by physical approximation or empirical calibration based upon long-term comparisons (Collier, 1996; Krajewski & Smith, 2002). As can be inferred, the parameters a and b are not constant but instead vary with DSD. In fact, disregarding DSD variability in space and time by adopting a fixed *Z-R* relationship is a considerable source of uncertainty in radar QPEs (Lee & Zawadzki, 2005).

In the case of polarimetric radars, additional variables are available (e.g., differential reflectivity [Zdr], differential phase [Kdp], and specific attenuation), which can be used for rainfall rate estimation. These variables provide information about raindrop shape and concentration, thus allowing dynamic adjustment of rainfall rate estimation, accounting for variations in drop size distribution (Bringi & Chandrasekar, 2001; Ryzhkov et al., 2014). At high rainfall rates the use of polarimetric estimators has been shown to improve radar rainfall estimation with respect to conventional estimators (Chen & Chandrasekar, 2015; Cifelli et al., 2011; Li & Mecikalski, 2012; Simpson & Fox, 2018). The benefits of polarimetric variables at lower rainfall rates are unclear, and in many cases the conventional *Z-R* approach leads to superior performance (Berne & Krajewski, 2013; Hagen & Meischner, 2000).

2.2.4. Uncertainties in Radar Rainfall Estimates

As has been widely discussed, there are large uncertainties associated with radar QPEs. The sources of these errors can be broadly categorized into (1) errors in return signal measurements, (2) errors in reflectivity-to-rainfall conversion, and (3) temporal sampling errors. The first two types are related to the indirect nature of radar rainfall measurements, while the last one is related to their representativeness.

Return signal measurements (i.e., reflectivity and additional parameters), from which radar QPEs are subsequently derived, may be affected by a variety of factors related to radar hardware, location in relation to target and atmospheric conditions. Errors in signal measurements include, among others, radar miscalibration, radar beam blockage, beam broadening, attenuation, ground clutter, anomalous propagation of the signal, and vertical profile variations (Collier, 1996; Einfalt et al., 2004; Harrison et al., 2000; Villarini & Krajewski, 2010). A number of corrections are usually applied in order to reduce errors arising from these sources (a summary of these corrections can be found in Table 1 of Harrison et al., 2009); however, it is virtually impossible to have error-free reflectivity measurements.

Additional errors arise from the conversion of received signal measurements (i.e., reflectivity [Z] and any other available parameters) to rainfall rates (R). As mentioned above, in the case of single polarization radars this is usually done using the *Z-R* relationship. The highly dynamic nature of rain drop size distribution, even within a single storm event (Smith et al., 2009; Ulbrich, 1993), renders the use of a static *Z-R* relationship—as used for single-polarization radars—imperfect, especially when extreme rainfall rates are observed (Einfalt, 2005; Goudenhoofdt & Delobbe, 2013; Villarini & Krajewski, 2010). In the case of polarimetric Doppler radars, additional information is collected, which enables significant reduction of some of the sources of error mentioned above (especially detection of nonweather echoes and improved attenuation correction), as well as a dynamic adjustment of the *Z-R* relationship according to drop-size distribution (Bringi et al., 2001; Bringi & Chandrasekar, 2001; Gu et al., 2011; Hagen & Meischner, 2000; Ryzhkov et al., 2014). This results in more accurate rainfall rate estimates, particularly at higher rain rates (Berne & Krajewski, 2013; Bringi & Chandrasekar, 2001; Hagen & Meischner, 2000). However, these advanced technologies will not change the existence of inherent limitations to radar as a rainfall measurement tool, such as the fact that rainfall is measured indirectly, often well above the ground and often far away from the radar.

The temporal sampling error in radar QPEs is associated with the instantaneous and periodic (as opposed to continuous) nature of radar measurements and the impact that this has on the estimation of rainfall accumulations, with the impact being particularly large when using high spatial-temporal resolution data

(Fabry et al., 1994; Seo & Krajewski, 2015). This error not only affects the accuracy of radar QPEs but also hinders the comparison of radar and rain gauge measurements; this in turn affects rain gauge-based radar calibration (Kitchen & Blackall, 1992) and radar-rain gauge merging (Anagnostou & Krajewski, 1999; Thorndahl et al., 2014; Wang et al., 2015). This error can be largely corrected by accounting for storm motion and/or evolution in the estimation of cumulative QPEs; for example, through temporal smoothing or advection-based correction (Fabry et al., 1994; Seo & Krajewski, 2015). The application of advection-based correction prior to merging has been shown to significantly improve the quality of the final QPEs (Anagnostou & Krajewski, 1999; Thorndahl et al., 2014; Wang, Ochoa-Rodríguez, van Assel, et al., 2015); this is discussed in more detail in section 6.2.

3. Radar-Rain Gauge Merging Methods

Numerous radar-rain gauge merging methods have been developed ever since radars started to be used as precipitation measuring devices. Likewise, different ways of categorizing existing merging methods have been proposed. Wang et al. (2013) suggest that merging methods can generally be classified as either bias reduction methods or error variance minimization methods; this classification was later adopted in the review by McKee and Binns (2015) and in Lo Conti et al. (2015). Erdin (2013), Goudenhoofdt and Delobbe (2009), and Jewell and Gaussiat (2015) make a clear distinction between geostatistical and nongeostatistical methods, where geostatistical methods are defined as those which seek to represent the spatial covariance structure of the rainfall field or its errors by making use of the (semi)variogram (Goudenhoofdt & Delobbe, 2009). The aforementioned categorizations focus mostly on the theories underlying the merging techniques. In contrast, Decloedt et al. (2013) proposed a more application-oriented categorization, which focuses on how each of the data sources (i.e., radar and rain gauge) are employed in the merging process. According to Decloedt et al. (2013), merging methods can be classified as either adjustment methods or integration methods. The former aim at adjusting radar estimates based upon rain gauge records, whereas the latter, as the name suggests, actually undertake a true integration of both data sources.

Given that the present review is application-oriented, a categorization similar to that of Decloedt et al. (2013) was adopted as starting point and it was further refined based upon the theoretical categorization by Wang et al. (2013). The proposed categorization is as follows: (1) methods focusing on bias adjustment of radar estimates; (2) rain gauge interpolation methods, which use radar spatial association as additional information; and (3) radar and rain gauge integration methods. As will be seen, methods in categories (2) and (3) generally fall within the error variance minimization type proposed by Wang et al. (2013).

A description of each of these categories and an overview of the techniques that belong to them is provided next.

3.1. Radar Bias Adjustment Methods

These methods attempt to correct the bias present in radar accumulations using rain gauge accumulations as the true rainfall value. The entire radar field is used as background and is simply adjusted by applying a multiplicative or additive correction factor. The correction factor is estimated based upon gauge-radar comparison (usually) at gauge locations over a given time period, which can either be long (i.e., nearly static adjustment) or short (i.e., dynamic adjustment; Borup et al., 2016; Fulton et al., 1998; Harrison et al., 2009; Hitschfeld & Bordan, 1954; Smith et al., 2007; Vieux & Bedient, 2004; Wilson, 1970; Wood et al., 2000b; Wright et al., 2014).

The simplest method in this category is the mean field bias (MFB) adjustment, which assumes that the radar estimates are affected by a (spatially) uniform multiplicative error due, for example, to an offset in the electronic calibration of the radar or in the *Z*-*R* relation used to convert radar reflectivity to rainfall rate. In this method, a multiplicative correction factor is applied uniformly across the radar domain. An example of the MFB method is presented in Figure 3. As can be seen, the MFB rainfall field corresponds to a simple scaling of the original radar field. The most common and simple method to estimate the MFB adjustment factor is as a spatially averaged ratio of rain gauge and co-located radar accumulations over the time period of choice (Harrison et al., 2009; Smith et al., 2007; Vieux & Bedient, 2004; Wilson, 1970). However, more elaborate methods have been developed for estimating the MFB correction factor, including maximum likelihood autoregressive algorithms (Smith & Krajewski, 1991); optimization methods, which aim at minimizing

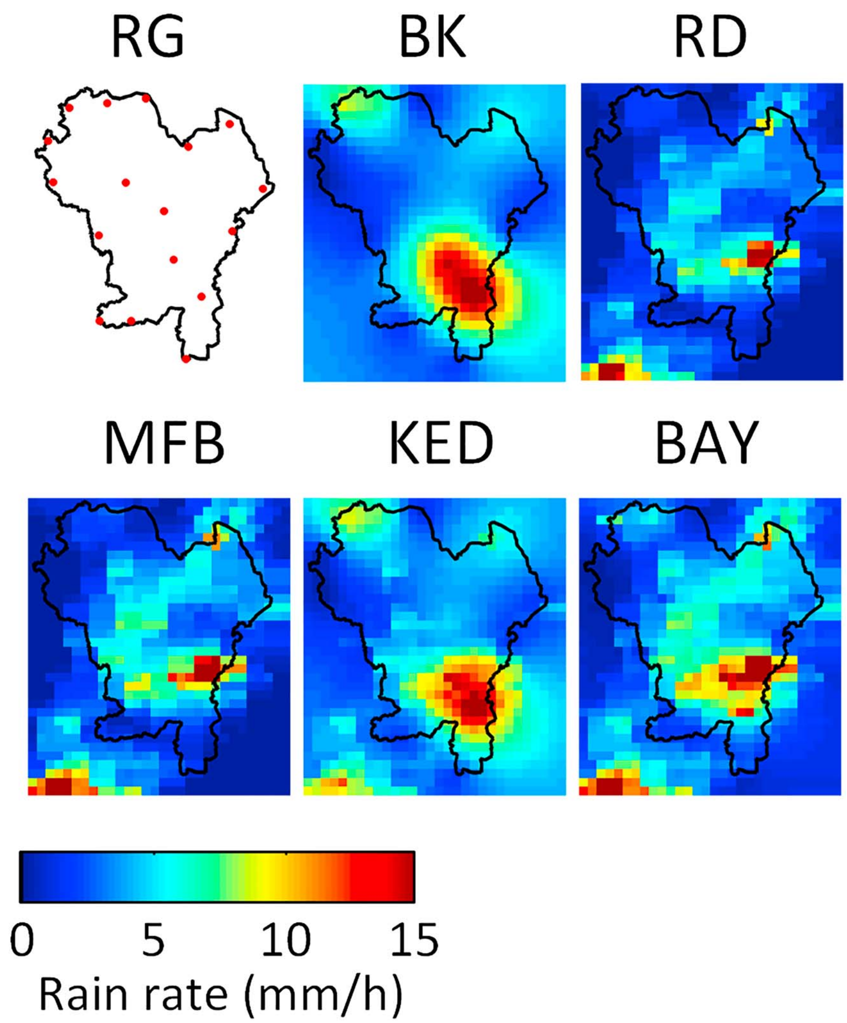

Figure 3. Example of merging results associated with merging methods belonging to each of the three categories defined in this paper: mean field bias (MFB) from the bias adjustment category, kriging with external drift (KED) from the rain gauge interpolation category, and Bayesian combination (BAY) from the integration category. Rain gauge (RG) location, block-kriged (BK) interpolated rain gauge records alone, and original radar (RD) records are shown in the top row. Radar records were available at 1-km^2/5-min resolution, while rain gauge records were available at 2-min temporal resolution. The results correspond to instantaneous rainfall rates across the Birmingham-Black Country area in the UK (30-km × 34-km area). Source: Ochoa-Rodriguez, Wang, Bailey, et al. (2015).

bias in antecedent time steps (Anagnostou et al., 1998; Anagnostou & Krajewski, 1999); Kalman filtering to account for bias in antecedent time steps (Anagnostou et al., 1998; Chumchean et al., 2006; Fulton et al., 1998), through linear regression analysis (Borup et al., 2016); and conditional on areal average rainfall rates (Thorndahl et al., 2014; Villarini et al., 2008; Wright et al., 2014). Recently, Biggs and Atkinson (2011) proposed a new method, which, instead of looking at radar-rain gauge differences exclusively at rain gauge locations, suggests estimating the MFB correction factor through comparison of the (co-kriged) interpolated rain gauge field and the radar field; this can help partially overcome rain gauge SSE and associated spatial sampling differences between rain gauge and radar. While this method entails geostatistical rain gauge interpolation prior to the correction, the actual correction falls within the bias adjustment category.

Unlike the MFB adjustment, other bias adjustment methods assume that radar bias changes in space and, as such, spatially varying bias corrections should be applied. For example, range-dependent bias correction assumes that radar bias increases with distance from the radar location as a result of beam broadening, overshooting, and attenuation effects (Michelson et al., 2000). Other methods, known as local bias correction methods, aim at deriving local correction factors through spatial interpolation or distribution of the radar-rain gauge differences at gauge locations. Different ways of interpolating these differences and deriving local correction factors have been devised. For example, Brandes (1975) proposed a negative exponential weighing method, based on Barnes's objective analysis, to derive multiplicative correction factors at each radar pixel. Similarly, Balascio (2001) and Cole and Moore (2008) proposed a multiquadratic surface fitting method to interpolate (modified) gauge-radar ratios and derive local multiplicative correction factors; this method was later modified by Martens et al. (2013) to make it more suitable for near-real-time applications. Another approach based on multiplicative bias correction was proposed by Seo and Breidenbach (2002): their method employs weighted least squares minimization to derive the expected gauge and radar rainfall estimate at a given location; the ratio of the expected values is used to correct the radar field. Pereira Fo et al. (1998) employed statistical objective analysis to derive additive bias corrections at each radar pixel based upon linear combination of errors at gauge locations; this technique was later modified by Gerstner and Heinemann (2008) for real-time applications. Other authors have simply used ordinary kriging (OK) to interpolate radar-gauge biases and associated correction factors (e.g., Erdin, 2013).

3.2. Rain Gauge Interpolation Methods Using Radar Spatial Association as Additional Information

Unlike bias adjustment methods, the methods in the interpolation category do not use the radar field as background; instead, they simply use the spatial association of the radar field to aid the spatial interpolation of point rain gauge values. The methods in the present category are all geostatistical and are extensions of kriging interpolation. The reader is reminded that kriging interpolation consists in predicting values at ungauged locations as the linear combination of values at gauged locations, with the linear weights being derived through minimization of the variance of the estimation error (Chiles & Delfiner, 2012). As such, kriging yields the best linear unbiased estimator of the rainfall field, based upon available rain gauge records.

The most widely used method in this category is kriging with external drift (KED; see Wackernagel, 2003). KED is an extension of kriging interpolation in which external variables, in this case radar QPEs, are used as auxiliary information in the interpolation process. In KED the linear weights employed in the interpolation of point gauge values are further constrained by the spatial association between radar values at the target and rain gauge locations. An example of KED merging results is presented in Figure 3. As can be seen, the KED field largely resembles the rain gauge-only interpolated (kriged) rainfall field, while displaying additional spatial features obtained from the radar field.

Note that KED and variations of it are widely used in other fields, besides radar-rain gauge merging (Chiles & Delfiner, 2012). A variation of KED, which has been recently applied for radar-rain gauge merging is indicator KED (Berndt et al., 2014; Haberlandt, 2007); indicator KED involves transformation of rain gauge data to binary data, based upon a number of thresholds (called “indicators”), prior to conducting the interpolation. More recently, another variation of KED, termed co-kriging (CoK) with external drift, has been proposed by Sideris et al. (2014). Their method applies radar as external information at each time step (as per standard KED), the difference lying in the introduction of rain gauge data from antecedent time steps as secondary CoK variable. As will be explained in the following section, CoK is considered an “integration” method; however, Sideris et al. (2014) did not employ CoK to merge radar and rain gauge data, but instead used it only to incorporate additional temporal information into the merging. As such, their method can be considered an extension of KED and is categorized as an interpolation method.

Another commonly used method in this category is conditional merging, also known as kriging with radar-based error correction (KRE). This method was formulated by Ehret (2002) and then implemented and tested by Sinclair and Pegram (2005). In KRE the radar field is used to characterize the features that cannot be well reproduced through kriging interpolation; this characterization is then used to correct the gauge-based kriged rainfall field. First, radar values at gauge locations are interpolated using OK; this yields a radar-based kriged field. This field is subtracted from the original radar field; the difference between the two fields is assumed to represent the errors associated with kriging interpolation. The error field is added to the gauge-based kriged field. In this way, the spatial structure of the radar field is incorporated into the gauge interpolated rainfall field.

As mentioned above, the methods in this category are geostatistical, kriging-based; as such, they all make use of the variogram to represent the spatial association of rainfall fields and one of their main underlying assumptions is that the rainfall field can be characterized as a Gaussian random variable. Because of this, the performance of these methods can be affected by the quality of the variogram, which in turn can be affected by the number and distribution of data samples available and by the degree of normality (or nonnormality) of the data. To deal with problematic cases in which data samples are too sparse to obtain a reliable variogram or in which the rainfall field displays significant non-Gaussian features, different variations of the aforementioned interpolation methods have been proposed. These include use of nonparametric variograms (Velasco-Forero et al., 2009), use of radar-based variograms (instead of gauge-based ones; Haberlandt, 2007), data transformation (Erdin, 2013), and decomposition (Wang et al., 2015) prior to interpolation, among others. It is worth noting that the purpose of the transformation and decomposition treatments is to improve compliance with Gaussian assumptions, as well as to preserve small-scale rainfall features, such as convective cells, which may be otherwise smoothed out during the interpolation-based merging process. These methodological choices are discussed in more detail in section 6.

3.3. Radar-Rain Gauge Integration Methods

The methods in this category do not use radar or rain gauge as background, but instead conduct an actual integration of both data sources with the final aim of minimizing the overall estimation uncertainty. In these methods, the rainfall value at a given location is estimated as the weighted average of radar and rain gauge values. While the formulation of integration methods may appear to be similar to that of some local bias adjustment methods, there are two main differences to be noted: (1) as the name suggests, local bias adjustment methods aim at minimizing local biases between radar and rain gauge estimates, whereas integration methods aim at minimizing overall estimation uncertainty (Todini, 2001), and (2) in local bias adjustment, radar is kept as background and the gauge estimates only change the local magnitude of radar precipitation, but they do not have a large impact on the shape of the radar field. In contrast, in integration methods, depending on their relative uncertainties, gauge and radar observations may override each other or simply be combined; in fact, in gauge-sparse regions the integrated rainfall field may be very similar to the radar field, whereas in regions with high density of gauges the final product is very similar to the gauge-only field (Kitzmiller et al., 2013).

There are two main methods in this category: CoK and the Bayesian (BAY) data merging as formulated by Todini (2001). While both methods aim at minimizing the estimation uncertainty, they do so in different ways. The CoK method incorporates both radar and rain gauge data into a single (co-) kriging system, and, by solving it, the variance of estimation errors is minimized. In contrast, Todini's BAY method explicitly quantifies the estimation uncertainty (in terms of estimation error covariance) of each data source and then combines them in such a way that the overall estimation uncertainty is minimized.

CoK is a multivariate variant of kriging, which enables the calculation of estimates for a poorly sampled variable (the primary variable; in this case gauge records) with the aid of a well-sampled variable (the co- or secondary variable; in this case, radar QPEs; Journel, 1989). The essential requirement for CoK is that the primary and secondary variables are highly correlated. Although CoK can be seen as a point interpolation method and its concept appears to be similar to that of KED—that is, both methods incorporate radar and rain gauge information into a kriging system—the fundamental difference between them lies in that, in KED, only the spatial association of radar data is used to constrain (by imposing large-scale variation into) the linear weights to be employed in the interpolation. In contrast, in CoK radar values actually contribute toward the final (merged) rainfall field and the merged field is a linear combination of both rain gauge and radar values. Several variations of CoK exist, with the main difference being the way in which the secondary variable is used. In Krajewski (1987) and Sun et al. (2000), the full form of CoK was used; this requires incorporation of the individual (auto-) covariances of rain gauge and radar, as well as their cross covariance, into the kriging system. This, however, results in a significantly larger kriging system, as compared to that of the original kriging or KED methods. In cases where the study domain is relatively big, such large kriging systems could be computationally expensive. Moreover, the large difference between the number of samples of the primary and secondary data often leads to a numerically unstable CoK matrix (close to singularity). To circumvent this problem, it is possible to use a reduced form of CoK, called (strictly) collocated co-kriging (CCoK; Chiles & Delfiner, 2012; Xu et al., 1992); examples of the application of this variation to rainfall merging can be found in Schuurmans et al. (2007) and Velasco-Forero et al. (2009). In the reduced CoK system, only the radar value at the target location is employed; consequently, only the cross covariance between all rain gauges and the radar value at the target location is incorporated into the kriging system (the radar auto-covariance is not needed). This can effectively reduce the size of the kriging system and stabilize the CoK matrix.

Unlike CoK, which conducts radar and rain gauge data combination into a single kriging matrix solution, Todini's BAY method includes three stages. First, the block-kriging technique is employed to interpolate the point rain gauge data into a field of rain gauge estimates with spatial resolution equal to that of the radar field. The associated estimation error covariance resulting from the kriging interpolation is used to represent the uncertainty of rain gauge data. Second, the interpolated rain gauge field is compared against the co-located radar field, and a field of radar-rain gauge differences can be obtained. The covariance of the resulting field is then used to derive the uncertainty of radar data. Lastly, a Kalman filter is employed to combine radar and rain gauge data at each location proportionally to their relative uncertainties, so that the overall estimation uncertainty is minimized. An example of BAY merging results is presented in Figure 3. As can be seen, the BAY field displays a mix of features from the two data sources. Note, for example, how more information from the radar field is visible in areas with lower rain gauge coverage, where the uncertainty (mainly the SSE) associated with rain gauge estimates is higher.

4. Studies Focusing on the Evaluation and Intercomparison of Radar-Rain Gauge Merging Techniques

The literature on the evaluation of radar-rain gauge merging methods and their application to hydrology is abundant. The general conclusion of the studies to date is that radar-rain gauge merging always leads to superior rainfall estimates as compared to radar QPEs alone and—almost always, except in cases of very dense rain gauge networks (Biggs & Atkinson, 2011; Schuurmans et al., 2007)—as compared to rain gauge records alone. While the vast majority of merging evaluation studies have focused on large-scale (often country-wide) applications, for which the spatial-temporal resolutions of interest are generally coarse (Anagnostou et al., 1998; Brandes, 1975; Cole & Moore, 2008; Fulton et al., 1998; Goudenhoofdt & Delobbe, 2009; Harrison et al., 2000; Harrison et al., 2009; Jewell & Gaussiat, 2015; Krajewski, 1987; Nanding et al., 2015; Rabiei & Haberlandt, 2015; Seo & Breidenbach, 2002; Sideris et al., 2014; Thorndahl et al., 2014), the applicability of these techniques at urban scales (and associated finer spatial-temporal resolutions) has increasingly gained interest in recent years (Borup et al., 2016; Looper & Vieux, 2012; Löwe et al., 2014; Smith et al., 2007; Thorndahl et al., 2014; Vieux & Bedient, 2004; Villarini et al., 2010; Wang et al., 2013). Available studies confirm the benefits of merging for both large and small-scale applications.

Despite the abundance of literature on this topic, most of the studies to date have focused on assessing the feasibility of a single merging method and comparing the performance of the resulting merged rainfall estimates against that of original radar and rain gauge estimates. Only few studies provide an intercomparison of various merging methods and, to the authors' knowledge, only two recent conference papers have provided a preliminary comparison of the three categories of merging methods discussed in the previous section (Kumar et al., 2016; Ochoa-Rodriguez et al., 2015). What is more, the existing intercomparison studies are almost exclusively focused on large-scale applications. As such, the question of the best performing and most suitable merging technique(s), particularly for urban applications, remains unanswered.

To address this question, in this section a review is conducted of studies focusing on the inter-comparison of merging methods, both at large and small scales. Based upon this, the best performing and most commonly used merging techniques of each type are identified.

Table 1 summarizes the findings of multiple studies focusing on intercomparison of merging methods, including details of the type of methods under investigation, the geographic extent of each study, the spatial-temporal resolutions at which merging was applied, and the method identified as best performing in the given study. Note that in most of these studies the performance of merging methods was assessed through comparison of merged rainfall estimates against rain gauge records, either through cross validation or based upon an independent network of rain gauges. While different features of the merging methods and associated rainfall estimates were assessed in the different studies, the overall best performance was

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Table 1
Review of Studies on Intercomparison of Radar-Rain Gauge Merging Methods, Including Categorization by Technique Type, Associated Geographic Scale and Spatial-Temporal Resolution of Application, and Summary of Best Performing Method Identified in Each Study

References	Merging method category			Geographic extent of application and rain gauge (RG) network employed in study	Spatial/temporal resolution	Best performing method(s)
	Bias adjustment	Interpolation	Integration			
Mazzetti and Todini (2004)	- Brandes - Modified Brandes (Koistinen & Puhakka, 1981)	-	- CoK - BAY	- Numerical experiment - 7-km × 7-km area - 9 pseudo RGs evenly distributed	1 km	BAY
Schuurmans et al. (2007)		- KED	- CCoK	- Three nested areas (225, 10,000, and 82,875 km^2) in the Netherlands - Combined RG data set from two networks: National RG networks (100 km^2/RG) and experimental RG network (7.5 km^2/RG)	2.5 km/1 day	KED
Goudenhoofdt and Delobbe (2009)	- MFB - Brandes - Range-dependent adjustment (RA) - Static kriged (local) bias adjustment + RA	- KED - KRE	-	- Meuse catchment (12,000 km^2) in Walloon region, Belgium - Merging (SPW) network: 160 km^2/RG - Verification (RMI) network: 110 km^2/RG	1.8 km/1 hr	KED
Velasco-Forero et al. (2009)	-	- KED	- CCoK	-A 19 600 km^2 area in Catalonia, Spain - SAIH RG network: 250 km^2/RG	1 km/1 hr	KED
Erdin (2013)	-	- KED - KRE	-	- Whole Switzerland (41,000 km^2) - MeteoSwiss RG network: (546 km^2/RG)	1 km/1 hr	KED
Wang et al. (2013)	- MFB	-	- BAY	- Cranbrook Catchment (9 km^2) in NE London, UK - ICL RG network: 3 km^2/RG	1 km/5 min	BAY
Berndt et al. (2014)	-	- KED - Indicator KED - KRE	-	- Hannover radar site (51,400 km^2), Germany - DWD RG network: 520 km^2/RG	1 km/10 min - 6 hr	KRE
Sideris et al. (2014)	-	- KED - Cokriging with External Drift (CKED)	-	- Whole Switzerland (41,000 km^2) - MeteoSwiss RG network: (546 km^2/RG)	1 km/10 min - 1 hr	CKED
Jewell and Gaussiat (2015)	- Multiquadratic Surface Fitting (MQ)	- KED - KRE	-	- England and Wales (151,000 km^2)\ - UKMO RG network: merging network (212 km^2/RG) and evaluation network (427 km^2/RG)	1 km/15 min - 1 hr	KED
Nanding et al. (2015)	- MFB	- KED - KRE	-	- North England (50,000 km^2) - EA RG network: merging network (300 km2/RG) and evaluation network (940 km^2/RG)	1 km/1 hr	KED
Ochoa-Rodriguez, Wang, Bailey, et al. (2015)	- MFB	-KED	- BAY - Singularity - sensitive Bayesian (sBAY)	- Minworth urban catchment (67 km^2), UK - RG network: 3.5 km^2/RG	1 km/5 min	KED and sBAY

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Table 1 *(continued)*

References	Merging method category: Bias adjustment	Merging method category: Interpolation	Merging method category: Integration	Geographic extent of application and rain gauge (RG) network employed in study	Spatial/temporal resolution	Best performing method(s)
Kumar et al. (2016)	- MFB - Range-dependent adjustment - Brandes - Kriged (local) bias adjustment	- KRE	- BAY - sBAY	- Upper Thames River Basin (3 482 km^2), Canada - RG network: 250 km^2/RG	Not specified	sBAY

Abbreviations: BAY, Bayesian; CCoK, collocated co-kriging; CoK, co-kriging; KED, kriging with external drift; KRE, kriging with radar-based error correction; MFB, mean field bias.

generally assessed based on accuracy measures, with the mean absolute error (MAE) being one of the most commonly used performance measures. Only few studies evaluated the performance of merging methods by comparing the hydrological outputs resulting from each rainfall input against depth and flow records (Kumar et al., 2016; Ochoa-Rodriguez, Wang, Gires, et al., 2015; Wang et al., 2013).

Based upon this review, the following observations can be made:

1. This review confirms that the vast majority of intercomparison studies have focused on large-scale (often country-wide) applications. In fact, many of the studies were conducted by National Weather Services (Erdin, 2013; Goudenhoofdt & Delobbe, 2009; Jewell & Gaussiat, 2015; Schuurmans et al., 2007; Sideris et al., 2014), whose interest lies in the overall performance of rainfall products across their country of jurisdiction. Given the large scales of interest, in these studies merging was conducted and evaluated at resolutions of the order of 1-3 km in space and hourly-daily in time, which are insufficient for urban drainage modeling applications (Ochoa-Rodriguez, Wang, Gires, et al., 2015). To the authors' knowledge, to date, only two intercomparison studies have focused on urban scales (Ochoa-Rodriguez, Wang, Bailey, et al., 2015; Wang et al., 2013); the former was limited to the evaluation of bias adjustment and integration methods based upon a network of only three rain gauges and the latter provides only preliminary insights into the relative performance of merging methods. While the findings of the large-scale studies are useful and provide insights into the general features of merging techniques, these findings may not be directly extrapolated to urban areas and further testing at smaller scales and associated spatial-temporal resolutions constitutes a research gap.
2. Bias adjustment and interpolation methods are much more widely used and tested than integration methods. This could be because integration methods are generally more complex in their formulation and implementation and are also more computationally demanding.
3. In general, interpolation and integration methods outperform bias adjustment methods (Goudenhoofdt & Delobbe, 2009; Jewell & Gaussiat, 2015; Kumar et al., 2016; Mazzetti & Todini, 2004; Nanding et al., 2015; Ochoa-Rodriguez, Wang, Bailey, et al., 2015; Wang et al., 2013). Note that that bias adjustment methods were not found to be the best performing ones in any of the intercomparison studies.
4. MFB is the most commonly used and investigated method within the bias adjustment category. While bias adjustment methods are generally outperformed by other methods, due to its simplicity MFB remains one of the most widely used merging methods for operational applications. In fact, it is the method currently used by many national meteorological services (Goudenhoofdt & Delobbe, 2016; Harrison et al., 2009; Thorndahl et al., 2013).
5. KED is the most popular and generally best performing interpolation method. In all studies under consideration, except for one, it was shown to outperform the KRE interpolation method (Erdin, 2013;Goudenhoofdt & Delobbe, 2009 ; Jewell & Gaussiat, 2015 ; Nanding et al., 2015). In the study by Berndt et al. (2014) KED was outperformed by KRE, with differences in performance being particularly evident at low rain gauge densities. However, most of the available evidence points toward KED as having an overall superior performance, as compared to that of KRE. In relation to noninterpolation methods, KED has been shown to outperform bias adjustment methods and also the CoK integration method (Schuurmans et al., 2007; Velasco-Forero et al., 2009). The only method that has displayed a similar or better performance than KED is the BAY (integration) method (Kumar et al., 2016; Ochoa-Rodriguez, Wang, Bailey, et al., 2015).
6. With regard to integration methods (namely, CoK and BAY), only few studies have assessed their performance against each other and against that of other types of methods. The available studies show that CoK-based methods are outperformed both by KED interpolation and BAY integration methods (Mazzetti & Todini, 2004; Schuurmans et al., 2007; Velasco-Forero et al., 2009). Furthermore, recent studies have highlighted the main shortcoming of CoK, which lies in the assumptions and approximations made to represent true (unknown) rainfall estimate uncertainty; this shortcoming may compromise the performance of this method and has led to a decrease in its use (McKee & Binns, 2015; Wang et al., 2013). The BAY integration method, on the other hand, has not only been shown to outperform CoK (Mazzetti & Todini, 2004) but has also been shown to outperform KRE (Kumar et al., 2016) and has displayed a similar performance to that of KED (Ochoa-Rodriguez, Wang, Bailey, et al., 2015).

7. Overall, the best performing methods are KED and BAY. Unfortunately, only one preliminary study (Ochoa-Rodriguez, Wang, Bailey, et al., 2015) has intercompared these two methods and has found their performance to be similar. Further investigation is needed to better understand the relative performance and features of these two methods.

Based upon this review and on the grounds of performance and wide-spread use, it is recommended that the MFB method from the bias adjustment category, KED from the interpolation category, and BAY from the integration category be intercompared in an urban context to provide conclusive evidence about their relative performance, ideally under a range of climatological and operational conditions, and thus provide guidance on merging application at urban scales.

5. Factors Affecting Merging Performance

Multiple factors, which affect the performance of radar-rain gauge merging techniques, have been identified and analyzed in the literature. However, as pinpointed by Erdin (2013), the most important single factor affecting the performance of radar-rain gauge merging methods is the ability of the two sensors to correctly and consistently sample the underlying weather conditions. The issue of consistency in the information provided by radar and rain gauges has increasingly gained attention in recent years and has been shown to have a profound impact on the performance of merging methods, particularly on those in the interpolation and integration categories (Erdin, 2013; Jewell & Gaussiat, 2015; Schuurmans et al., 2007; Wang, Ochoa-Rodríguez, Onof, & Willems, 2015), with the best merging results obtained when the measurements from radar and rain gauges are consistent.

This is illustrated in Figure 4, adapted from Schuurmans et al. (2007), which depicts the ratio of the root-mean-square error of the OK rainfall product and that of the merged rainfall products (KED and CCoK) as a function of the correlation between rain gauge and co-located radar measurements. Results are presented at control rain gauge locations (following a cross-validation approach) across three geographic domains, respectively of 225, 10,000, and 82,875 km^2 across the Netherlands. Root-mean-square error ratios greater than 1 indicate better performance of merged products in relation to the OK (rain gauge-only) product, thus providing an indication of the added value of radar incorporation via merging. As can be seen, merging performance (and the associated added value of merging) generally increases with increasing correlation between rain gauge and radar measurements; this is particularly evident for medium to large geographic domains, but is still visible for the smaller domains.

This sampling and consistency factor is, in turn, a function of multiple and interacting subfactors, which can be broadly categorized as climatological and operational factors. An overview of these subfactors and a review of studies focusing on their impact and handling are provided next.

5.1. Climatological Factors

Climatological factors have to do with climatic and storm characteristics in the area of interest, which are in turn determined by geographic location, local topography, and seasons, among other features. Several studies suggest that storm type—which is highly related to seasons—is one of the main factors affecting the performance of merging methods (Erdin, 2013;Goudenhoofdt & Delobbe, 2009 ; Jewell & Gaussiat, 2015 ; Nanding et al., 2015). In these studies, storms were generally categorized into stratiform (winter) and convective (summer) types. Stratiform storms are characterized by relatively lower rainfall rates, lower spatial variability, and large characteristic lengths (i.e., covering large spatial extents). Convective storms, on the other hand, are characterized by higher rainfall rates and spatial variability, by having a smaller extent (of the scale of ~10 or more kilometers) and by displaying a higher degree of nonlinearity (i.e., non-Gaussian behavior; Hubert et al., 1993; Lovejoy & Mandelbrot, 1985; Lovejoy & Schertzer, 1995; Schertzer & Lovejoy, 1987; Wang et al., 2012). Because of the spatial features of stratiform events, the spatial sampling density required to effectively capture them is generally low. In fact, rain gauge records alone often provide a satisfactory representation of these storms and the added value of radar is often limited in the case of these events (Goudenhoofdt & Delobbe, 2009). Conversely, the correct capturing of convective storms requires higher spatial sampling densities, which are hard to achieve with operational rain gauge networks. As a result, it is in the case of convective storms that the added value of radar data is truly evident, as radar QPEs can compensate for the insufficient spatial information provided by gauges (Goudenhoofdt & Delobbe, 2009;

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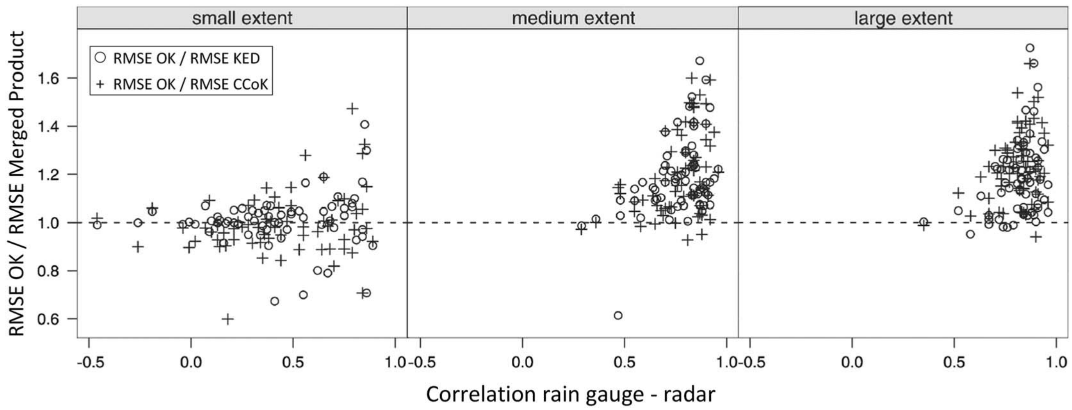

Figure 4. Impact of radar-rain gauge correlation on merging performance: ratio of root-mean-square error (RMSE) of ordinary kriging (OK) rainfall product and that of the merged rainfall products (kriging with external drift [KED] and collocated co-kriging [CCoK]) as a function of the correlation between rain gauge and radar measurements. Results are presented for three geographic domains, respectively of 225, 10,000, and 82,875 km^2 across the Netherlands. Analyses were conducted at 2.5-km/1-day resolution. Source: Schuurmans et al. (2007).

Schuurmans et al., 2007). In effect, in the case of convective events, merged rainfall products almost always significantly outperform rain gauge and radar only rainfall products. The aforementioned features of the two storm types and the ability of rain gauges and radar to sample them also impact the consistency in the information of the rainfall field provided by the two sensors. In the case of stratiform storms, gauges and radar generally provide a consistent picture of the rainfall field. On the contrary, the high spatial (and associated temporal) variability of convective storms, along with the higher rainfall rates associated with these events, which are often inaccurately captured by radar QPEs (Einfalt et al., 2005), frequently lead to large differences between gauge and co-located radar records and between the rain gauge-based (interpolated) rainfall field and the radar image; this is particularly the case when data are analyzed and employed at fine temporal resolution (Seo & Krajewski, 2015; Sideris et al., 2014; Thorndahl et al., 2014).

The better sampling of stratiform storms by gauges and the better consistency between radar and rain gauge records for such events leads to better overall performance (in terms of accuracy) of merging methods for these events, as compared to their performance in the case of convective storms (Goudenhoofdt & Delobbe, 2009; Jewell & Gaussiat, 2015; Nanding et al., 2015). However, as mentioned before, it is in the case of convective storms that the added value of merging, in particular of using radar measurements, is more evident. An example from Nanding et al. (2015) is presented in Figure 5. This figure illustrates the performance of different merging products and of original radar and interpolated rain gauge estimates under different storm types. As can be seen, merging performance is generally better for stratiform storms than for convective ones. However, the added value of merging is more significant under convective storms (e.g., notice the better performance of KED as compared to original radar and interpolated rain gauges estimates under convective storms, particularly for the higher rainfall threshold).

Obviously, climatological factors cannot be “manipulated” and all that can be done is to understand and quantify their impact by testing merging methods under different climatological conditions. There are, however, ways of treating the radar and rain gauge data prior to and during merging, so that the complications associated with climatological factors, particularly those associated with convective storms, can be better dealt with and that the overall performance of merging methods is improved. These treatments fall within the operational factor category and are addressed in the following section.

5.2. Operational (Infrastructural) Factors

Operational infrastructural factors are related to the existing radar and rain gauge network infrastructure in the area of interest and are either permanent or difficult to modify in the short term by hydrologists, particularly by urban hydrologists.

In the case of radar, infrastructural factors include radar location in relation to the catchment (see Figure 6a), radar hardware, scanning strategy, and QPE generation algorithms, including signal

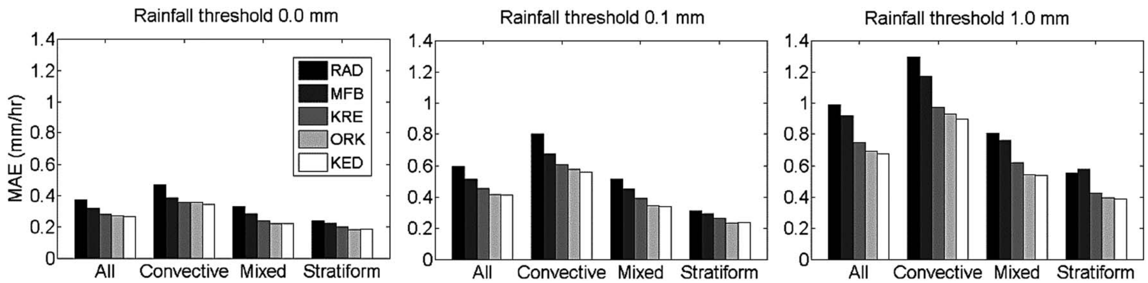

Figure 5. Impact of storm type on merging performance: mean absolute error (MAE) of hourly radar (RAD), ordinary kriging interpolated rain gauge data (ORD), and merged estimates (mean field bias [MFB], kriging with radar-based error correction [KRE], and kriging with external drift [KED]) at control gauge locations under different storm types. Source: Nanding et al. (2015).

correction and reflectivity conversion algorithms (see section 2.2). While QPE generation algorithms can perhaps be regarded as a methodological choice, in the context of this study they will be considered as an infrastructural factor given that the QPE generation process (from raw radar measurements) is seldom under the control of the urban hydrologist but is instead usually controlled by national meteorological services.

In the case of rain gauges, infrastructural factors include type and precision of the rain gauges, number, and location of gauges in the area of interest (i.e., gauge density and network configuration), and existing logging and transmission system. These infrastructural factors determine the accuracy and spatial-temporal sampling of radar QPEs and rain gauge estimates, which in turn condition the final quality and resolution of merging estimates.

One of the most important infrastructural factors, which has been analyzed in multiple radar-rain gauge merging studies, is rain gauge density. Gauge density determines the spatial sampling of storms by gauges and has been shown to have a strong impact on the performance of merging methods, particularly of those within the interpolation category and in the case of convective storms (Goudenhoofdt & Delobbe, 2009; Jewell & Gaussiat, 2015; Nanding et al., 2015). This is illustrated in Figure 6b); as can be seen, rain gauge density has a clear impact on merging performance, particularly for interpolation-based merging methods such as KED and KRE. Unfortunately, studies focusing on the impact of rain gauge density on merging performance and on the identification of gauge density requirements have been limited to large scale

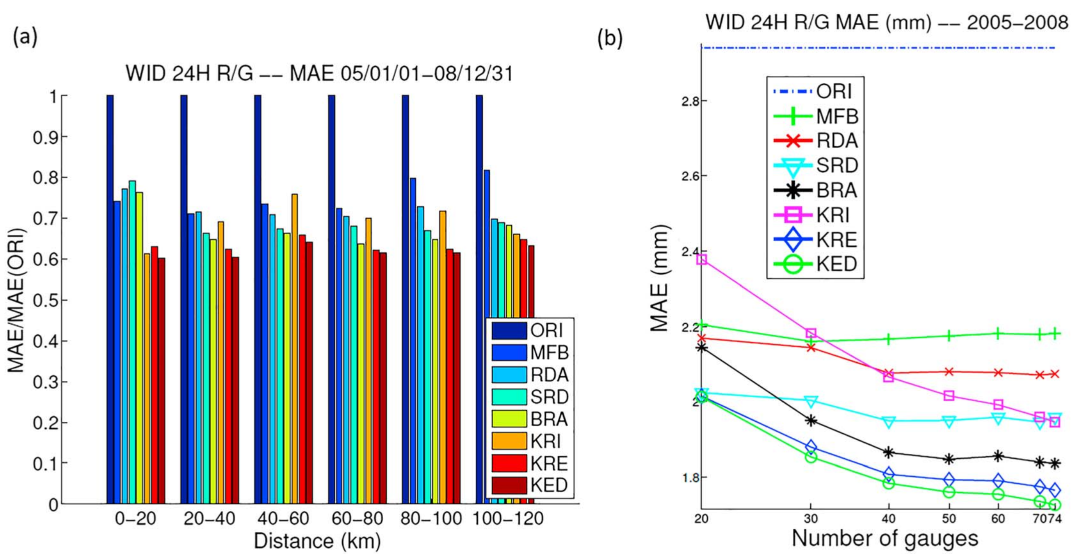

Figure 6. Impact of infrastructural factors on merging performance (at 1-hr/1.8-km resolution): (a) merging performance as a function of distance from radar; (b) merging performance as a function of gauge network density. Rainfall products: original radar data (ORI), mean field bias adjustment (MFB), range-dependent adjustment (RDA), statistical local bias correction and range-dependent adjustment (SRD), Brandes spatial adjustment (BRA), ordinary kriging (KRI), kriging with radar-based error correction (KRE), kriging with external drift (KED). Source: Goudenhoofdt and Delobbe (2009).

applications. To the authors' knowledge, only few preliminary analyses have been reported of the impact of gauge density on merging outputs at urban scales (Ochoa-Rodriguez, Wang, Bailey, et al., 2015); a likely reason for the lack of such studies at urban scales is that the number of gauges available in urban catchments is generally insufficient to enable a gauge density analysis.

6. Methodological Choices Affecting Merging Performance

Methodological choices relate to how rain gauge and radar data are treated prior to merging and how merging is applied. Methodological choices are often tailored to handle the impact of or adapt to some of the climatological and operational factors discussed in the previous section. In many cases, methodological choices aim at improving the quality of, and consistency between, radar and rain gauge estimates, which, as discussed in the previous section, is the most important single factor affecting the performance of radar-rain gauge merging methods. An overview and discussion of critical methodological choices, several of which relate to the application of merging at fine spatial-temporal scales, is given next.

6.1. Rain Gauge Quality-Control Prior to Merging

The main treatment to which rain gauge records can be subjected prior to merging and which is widely applied is quality-control (Gjertsen et al., 2003; Jewell & Gaussiat, 2015). Quality control of rain gauge records can be performed based on fixed rules or thresholds (e.g., to check for physically unlikely records), through comparison against neighboring values (also known as spatial consistency checks) and against co-located radar estimates (e.g., based upon correlation analysis). Quality-control of gauge records enables identification and eventual correction of erroneous measurements, thus improving the accuracy of the gauge estimates. Given that one of the main assumptions of merging methods is that point rain gauge records are reliable and that these can be used to correct radar rainfall rates, any efforts made to improve the accuracy of gauge estimates can effectively improve the quality of the final merged product (Erdin, 2013; Gjertsen et al., 2003).

6.2. Radar Quality-Control and Correction Prior to Merging

Radar QPEs can also be subjected to quality-control prior to merging (e.g., Rabiei & Haberlandt, 2015); in fact, some merging methods, such as the BAY combination (Todini, 2001), include in their formulation bias correction of radar QPEs prior to merging. However, the extent to which radar QPEs can be quality-controlled and improved prior to merging is limited (only very simple methods, such as bias correction, have been reported in the literature) and the purpose of merging is, in fact, to correct errors in radar QPEs based upon rain gauge records. Other treatments to which radar QPEs can be subjected prior to merging include advection correction and temporal interpolation of radar images (Anagnostou & Krajewski, 1999; Fabry et al., 1994; Seo & Krajewski, 2015; Thorndahl et al., 2014; Wang, Ochoa-Rodríguez, van Assel, et al., 2015), spatial and/or temporal smoothing (Berndt et al., 2014), data decomposition (Wang, Ochoa-Rodríguez, Onof, & Willems, 2015), and transformation (Berndt et al., 2014; Cecinati et al., 2017; Erdin, 2013).

The aim of advection correction and temporal interpolation of radar images is to reduce the temporal sampling errors in radar QPEs, as well as the differences in radar and rain gauge estimates arising from the instantaneous versus cumulative nature of the measurements, respectively. Advection correction of radar images was effectively used by Anagnostou and Krajewski (1999) to reduce radar temporal sampling error prior to conducting MFB adjustment. Similarly, advection-based temporal interpolation of radar images has been effectively used by Thorndahl et al. (2014) prior to applying MFB merging and by Wang, Ochoa-Rodríguez, van Assel, et al. (2015) prior to applying BAY merging. In all cases, reduction of radar temporal sampling error has been shown to improve the applicability of radar QPEs for hydrological applications and has led to better merging results. An example is given in Figure 7, where the performances of MFB-adjusted rainfall estimates resulting from the original and from the advection-based interpolated radar rainfall estimates are compared. As can be seen, the corrected radar QPEs result in improved merging performance (see higher Nash-Sutcliffe efficiency coefficient and lower MAE for the corrected radar product).

It is worth mentioning that radar advection correction and temporal interpolation treatments can be particularly useful for hydrological applications requiring high temporal resolution rainfall data (Wang, Ochoa-Rodríguez, van Assel, et al., 2015), given that it is at these fine scales at which temporal sampling errors and

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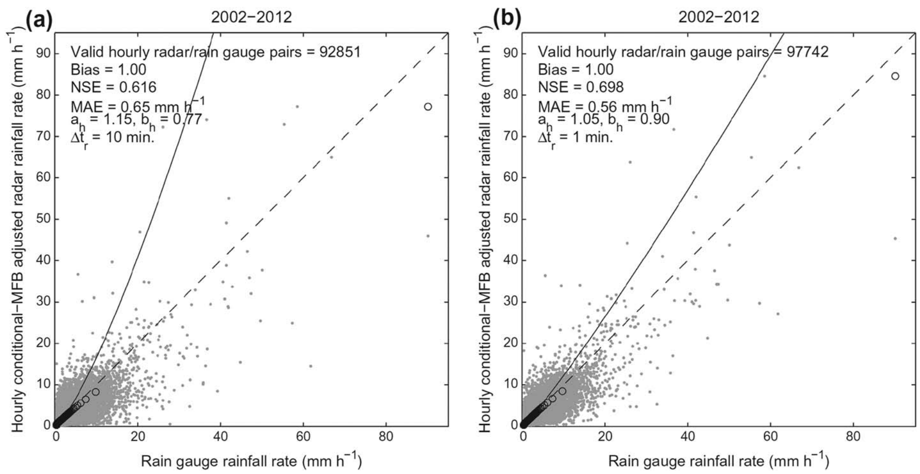

Figure 7. Impact of advection-based radar correction on merging performance. Scatterplots of MFB adjusted rainfall rates versus rain gauge rainfall rates at hourly time scale: (a) mean field bias (MFB) adjustment based on original radar data, prior to advection-based correction; (b) MFB adjustment based on advection-based temporally interpolated radar data. Statistics of interest: Nash-Sutcliffe efficiency coefficient (NSE) and mean absolute error (MAE). Source: Thorndahl et al. (2014).

discrepancies between radar and rain gauge measurements are more evident (Ochoa-Rodriguez, Wang, Gires, et al., 2015;Schuurmans et al., 2007 ; Sideris et al., 2014).

6.3. “Normalization” of Radar Data Prior to Merging Through Transformation and Decomposition

The purpose of radar data transformation and decomposition treatments is to “normalize” the radar field (i.e., to smooth off nonlinearities, which are especially common in the case of convective storms), so that better compliance with the assumptions of some merging methods (particularly those that rely on geostatistical modeling) is achieved and that a better correlation can be attained between radar and rain gauge records. The impact of radar data transformation on geostatistical merging methods (both within the interpolation and integration categories) has been tested by Schuurmans et al. (2007), Erdin et al. (2012), Erdin (2013), and Cecinati, Wani, and Rico-Ramirez (2017). In these studies, Box-Cox (including logarithmic), square-root and normal score data transformations were tested alongside KED, KRE, and ordinary CoK merging methods. Said transformations are applied to the radar and rain gauge records prior to merging, and a back-transformation is performed after merging is applied. In general, it was found that radar data transformation leads to better compliance with geostatistical modeling assumptions and this sometimes leads to improvements in the reliability of merged estimates (i.e., estimation uncertainty was effectively reduced). However, in most cases, particularly in the case of the Box-Cox transformation, improvements were found to be inconsistent and the optimal transformation function (i.e., that leading to best performance) was found to be strongly event-dependent (Cecinati, Wani, & Rico-Ramirez, 2017; Erdin, 2013). This may hinder the use of this treatment for operational applications. It is worth noting that the normal score transformation is less event dependent and was found to lead to overall improvements in KED merging performance by Cecinati, Wani, and Rico-Ramirez (2017). However, its application to rainfall data is rather complex due to the discontinuous and nonmonotonic nature of the data. In addition, the back transformation is computationally slow as it requires linear interpolation/extrapolation for a sufficient number of quantiles for each pixel. In addition to these complications, radar data transformation approaches have only been tested at larger scales and coarse temporal resolutions (of 1 hr (Cecinati, Wani, & Rico-Ramirez, 2017; Erdin, 2013) and 1 day (Schuurmans et al., 2007)). Their applicability at the high temporal resolutions required for urban hydrological applications (of the order of 1-5 min) may further complicate things due to the higher intermittency in rainfall values (i.e., many zeroes); this could be particularly critical in the case of the logarithmic transformation. More recently, a data decomposition technique, based upon local singularity analysis, was proposed and tested by Wang, Ochoa-Rodríguez, Onof, and Willems (2015). This technique aims at improving interpolation and integration methods involving geostatistical modeling by (1) improving the compliance of radar rainfall fields with model assumptions and (2) ensuring preservation of fine-scale singularity structures (local non-Gaussian features of the rainfall field), which have been found to be smoothed out by geostatistical merging methods. The distribution of singularity structures is highly consistent with that of local extremes and fine-scale rainfall features (Wang et al., 2012), hence the importance of preserving them, particularly for applications requiring high spatial resolution rainfall estimates. The proposed technique consists in decomposing the radar field prior to merging by removing from it the so-called singularities. This results in a nonsingular radar rainfall field, which is smoother, more normal, and which is also more likely to display a better correlation with rain gauge-based rainfall fields. The nonsingular radar rainfall field is merged with rain gauge records. Afterward, the singularity structures are reintroduced proportionally to the merged rainfall field, thus ensuring their preservation throughout the merging process. This decomposition method was initially tested in combination with the BAY merging method in an urban context and was shown to effectively improve merging results (Wang, Ochoa-Rodríguez, Onof, & Willems, 2015). Later on, and this time in a larger geographic domain (River Thames catchment, Canada), Kumar et al. (2016) compared the performance of the BAY method with singularity decomposition treatment against that of other merging methods (including MFB, KRE and Brandes) and found the singularity-BAY method to outperform others. However, Kumar et al. (2016) did not specifically analyze the impact of the singularity treatment on merging performance: they simply applied the singularity-BAY method but did not compare its performance against that of the original BAY method, without singularity treatment. More recently, Cecinati, Wani, and Rico-Ramirez (2017) tested the singularity decomposition approach in combination with KED, with merging performed over a large geographic domain (a 200-km × 200-km area in the North of England) and at hourly temporal resolution. The associated results confirm that the singularity decomposition treatment can help retain small-scale features observed in the original radar images. However, improvements in actual accuracy were not tangible. The ability of the singularity data decomposition treatment to preserve small-scale features may be particularly useful for urban applications; nonetheless, its application at urban scales has so far been limited to the BAY integration method (Wang, Ochoa-Rodríguez, Onof, & Willems, 2015). The benefits of its application in combination with other merging methods (different from BAY integration) at urban scales remains to be tested, as well as its suitability when applied in combination with other radar data treatments, such as temporal interpolation or advection correction.

6.4. Spatial Smoothing of Radar Data Prior to Merging

Spatial smoothing of radar data (e.g., reestimation of radar grid value as the average of neighboring pixels) has a similar aim and impact as those of the data decomposition and transformation treatments, while also reducing the noise-to-signal ratio in radar estimates. Berndt et al. (2014) applied several radar spatial smoothing methods prior to conducting KED and KRE merging. They concluded that smoothing of the gridded radar data significantly improves the performance of KED and KRE merging methods, but it also leads to loss of variability in rainfall estimates. Such loss of variability can be acceptable for large-scale hydrological applications but is likely unsuitable for urban drainage modeling applications.

6.5. Temporal Resolution at Which Merging is Applied

A critical methodological factor, which has been shown to have a profound impact on merging results, is the temporal resolution at which merging is conducted. The correlation between radar and rain gauge records has been found to increase as the integration time increases (Schuurmans et al., 2007; Sideris et al., 2014); consequently, better merging results are obtained when merging is applied at coarser temporal resolutions (Berndt et al., 2014; Cecinati et al., 2017; Goudenhoofdt & Delobbe, 2009; Jewell & Gaussiat, 2015). For example, Jewell and Gaussiat (2015) reported a better performance of KED and KRE methods when applied at a temporal resolution of 1 hr, in relation to their performance at a temporal resolution of 15 min. In their study on the impact of radar data smoothing, Berndt et al. (2014) also tested the performance of KED and KRE methods at temporal resolutions ranging from 10 min to 6 hr. In general, lower errors and better merging performance were observed at coarser temporal resolutions; however, the authors demonstrated that merging can lead to improvements over radar and rain gauge only estimates even at the finest temporal resolutions. As has been highlighted in multiple occasions, using coarser temporal resolution rainfall estimates is not an option for urban hydrological applications; as such, other ways of improving the performance of merging methods, different from increasing the temporal resolution at which merging is performed, should be sought by urban hydrologists. It is worth noting that Sideris et al. (2014) suggested a potential workaround for cases in which high temporal resolution rainfall estimates are needed: they propose generating 1-hr merged QPEs and then disaggregating to finer temporal resolutions (e.g., 5 min) based on the temporal profile of the 5-min original radar data. This approach was not tested by Sideris et al. (2014) but was instead recently tested by Cecinati, de Niet, et al. (2017) in an urban context. Cecinati, de Niet, et al. (2017) undertook rainfall aggregation at various temporal resolutions, from 1 day to 1 hr, and performed KED merging at these resolutions; the merged estimates were then downscaled to three finer resolutions (of 5, 15, and 30 min) based on the temporal profile of the radar time series. The method was expanded to generate probabilistic merged-downscaled estimates. The resulting merged-downscaled rainfall estimates were used as input to the urban drainage model of a Dutch catchment for which water-level records were available at one location. For this specific case study, best results were achieved for the rainfall product accumulated at 3 hr for KED application and then downscaled to 30 min. While this methodology is interesting and promising, the validation presented by Cecinati, de Niet, et al. (2017) is limited to a single and rather atypical catchment (characterized by being flat—as is typical of Dutch catchments—and containing a high proportion of pumped flows) and has several shortcomings, one of the main ones being that merged products resulting from original radar and rain gauge estimates at fine temporal resolutions (e.g., 5 min, without application of downscaling) were not included in the assessment. Likewise, the identified optimal temporal resolution (i.e., 30 min) is well coarser than the critical temporal resolution for urban sewer modeling identified in the multistorm, multicatchment investigation undertaken by Ochoa-Rodriguez, Wang, Gires, et al. (2015). Further testing of this approach is required to validate the findings and identify general optimal aggregation and downscaled (final) temporal resolutions.

6.6. Choice of Variogram (Geostatistical Merging Methods)

Another implementation factor, which has been widely analyzed, is the way in which the variogram to be used in geostatistically based interpolation (e.g., KED and KRE) and integration methods (e.g., CoK and BAY) is derived (Erdin, 2013; Schuurmans et al., 2007; Velasco-Forero et al., 2009). The variogram to be used in these methods can be either derived based on rain gauge records or on radar QPEs and can be modeled using parametric or nonparametric variogram approaches. While the nonparametric variogram approach can be computationally more efficient, it has been found to lead to overly smooth rainfall fields. In any case, the way in which the variogram is derived has been found to bear little impact on overall merging results (Erdin, 2013; Schuurmans et al., 2007).

6.7. Handling of Rain Gauge Records in Areas of Convection

Lastly, another implementation choice worth mentioning is the handling of rain gauge records in areas of convection during the merging process. As discussed in section 5, radar and rain gauges often provide inconsistent sampling of convective storms and this can significantly affect the performance of merging methods. To solve this problem, two recent studies have suggested a similar approach consisting in, for each time step, identifying areas of convection (based on an analysis of surrounding rainfall intensities) or areas in which rain gauge—radar correlation is low and excluding the records from gauges located in these areas from the merging (Jewell & Gaussiat, 2015; Laurantin, 2013). This approach has reportedly led to overall improvements, at the national (England) level, in KED performance under convective conditions (Jewell & Gaussiat, 2015). However, such exclusion of rain gauge records from the merging process could lead to loss of valuable information, particularly about high rainfall rates often associated with convective storms and which are of utmost importance for hydrological applications. Furthermore, in the case of urban applications, where the number of gauges available is usually small (of the order of tens and often less than 10; e.g., Borup et al., 2016; Notaro et al., 2013; Pina et al., 2016; Wang et al., 2013), excluding some gauges from merging is simply unfeasible.

7. Discussion: Potential of Merging Methods for Urban Hydrological Applications

The potential of a merging method for a given hydrological application is a function of multiple factors, including the quality of the underlying radar and rain gauge records, the way in which the data are treated prior to merging, the climatological, and operational (infrastructural) conditions in the catchment(s) of interest, and the specific requirements of the application at hand (e.g., required spatial-temporal resolution of rainfall estimates, whether merged rainfall estimates are required in real time or not, available computational resources, among others).

To explore the potential of the different merging methods (as per categories defined in section 3) for the urban hydrological applications of interest, in this section a conceptual intercomparison of the different merging methods is presented. This intercomparison draws upon the findings of the previous sections and looks at the relative strengths and weaknesses of each of the methods, with a focus on the way in which they handle the specific challenges associated with the application of radar-rain gauge merging at urban scales. These include the following:

1. Measurement uncertainties and spatial-temporal sampling differences in radar and rain gauge measurements are generally magnified at small scales (Ciach & Krajewski, 1999a; Villarini, Mandapaka, et al., 2008). As discussed in sections 5 and 6, this can have a knock-on effect on merging performance. It is therefore vital to understand how different measurement uncertainties can be handled prior to and throughout the merging process.
2. The preservation of small-scale, non-Gaussian rainfall features is critical for urban applications, particularly as their distribution is highly consistent with that of local extremes, including those associated with convective storms (Tchiguirinskaia et al., 2011; Wang et al., 2012). Such features are however often smoothened or insufficiently corrected in the merging process. While this may be acceptable for larger scale applications, it is certainly not the case for the stringent urban applications of interest.
3. Ground rainfall monitoring in urban areas is a challenging task, due to the high density of buildings (mostly privately owned and not accessible to environmental or water resources agencies) and limited availability of open spaces where the rainfall field is less obstructed (Chapman et al., 2015; de Vos et al., 2017). As a result, the number of available rain gauges over urban catchments is often small. This can significantly increase the uncertainty associated with using rain gauge data as ground truth (or unbiased rainfall estimates) in the merging process, with the impact varying across merging methods depending on their reliance upon rain gauge data and on the proportion of radar data used in the merging process.
4. Performing merging at fine spatial-temporal resolution results in increased computational requirements and runtimes. This challenge may be further exacerbated by the increasing availability of higher spatial-temporal resolution radar products both from national and local radar networks (Chandrasekar et al., 2009; Mishra et al., 2016; Ochoa-Rodriguez et al., 2015b; Yoon & Nakakita, 2017) and of ground rainfall measurements from alternative sources, such as citizen observatories (de Vos et al., 2017; Muller et al., 2015).

A discussion of each of these challenges and their handling throughout the merging process is presented in the next sections. While these challenges are inter-related, for clarity purposes each of them will be discussed separately. At the end of this section a summary intercomparison of the three merging methods is presented.

7.1. Handling of Measurement Uncertainty Throughout the Merging Process

As discussed in section 2, rain gauge and radar measurements are subject to errors of different types. As discussed in section 6, some of these errors can and should—at least partially—be corrected prior to merging; this is, for example, the case of rain gauge systematic and operational errors (see section 6.1), and of all three types of radar errors (see sections 2.2 and 6.2). In particular, in the case of radar data, the correction of temporal sampling errors prior to merging has been shown to increase the correlation of radar and rain gauge measurements, which in turn has led to significant improvements in merging results. This can be particularly useful for hydrological applications requiring high temporal resolution rainfall data (Wang, Ochoa-Rodríguez, van Assel, et al., 2015), given that it is at these fine scales at which temporal sampling errors and discrepancies between radar and rain gauge measurements are more evident (Ochoa-Rodriguez, Wang, Gires, et al., 2015; Schuurmans et al., 2007; Sideris et al., 2014).

However, some measurement errors can either not be treated, or not be satisfactorily corrected prior to merging; for example, random rain gauge errors, rain gauge SSEs, and residual errors in radar QPEs. It is desirable to account for this type of error throughout the merging process, particularly when dealing with urban-scale applications. Considering this, it is vital to examine, from a methodological point of view, if and how different merging methods allow for the incorporation of individual measurement uncertainties into the merging process.

Before delving into this topic, it is worth noting that, strictly speaking, it is methodologically possible for all three types of merging methods to incorporate a statistical model to account for radar and/or rain gauge measurement uncertainty, based upon which ensemble of realizations can be stochastically generated; this however falls within the realm of rainfall error modeling (AghaKouchak et al., 2010; Delrieu et al., 2014; Germann et al., 2009; Ochoa-Rodríguez et al., 2016; Rico-Ramirez et al., 2015), which is outside of the scope of this paper. Instead, the focus here lies on analyzing if and how a given merging method allows for explicit incorporation of measurement uncertainty throughout the merging process, which could directly impact the (deterministic) merged estimate. It is worth noting that to the authors' knowledge, there is not much information on this topic in the literature; as such, the following discussion is based upon our review of the mathematical formulation of the different merging methods.

In general, bias adjustment methods (e.g., MFB) do not intrinsically enable the incorporation of measurement uncertainty into the merging process. This is mainly because most methods in this category assume rain gauge records to be unbiased rainfall estimates and characterize the difference between radar and rain gauge data based upon a first-order statistical model, so the incorporation of measurement uncertainty does not lead to immediate changes in the resulting merged rainfall estimates. Some of the more sophisticated bias adjustment methods may however characterize radar-rain gauge biases by means of higher-order models; this could result in changes to the merged estimates. For example, the application of kriging interpolation to rain gauge data or to the difference between rain gauge and co-located radar measurements prior to undertaking bias adjustment would—in theory—enable the incorporation of rain gauge random error (Ciach, 2003; mainly in the form of variance) into the process of generating spatially distributed biases (for information on the incorporation of measurement uncertainty into kriging interpolation the reader may refer to Mazzetti, 2004). However, to the authors' knowledge, measurement uncertainty is generally not considered in the application of bias adjustment methods.

Similar to the more sophisticated bias adjustment methods, interpolation-based methods, which are mostly developed on the basis of the theory of kriging, make it possible to account for rain gauge random error in the form of variance; this has the potential to change the final merged estimates. However, to the authors' knowledge, this approach is generally not adopted in the application of interpolation-based merging methods. As regard rain gauge SSE, this is not accounted for in the majority of interpolation-based methods, except for KRE. The formulation of the KRE method entails estimation of rain gauge SSE through comparison of the original radar field against the field resulting from kriging interpolation of radar measurements at rain gauge location. As regard radar uncertainty, methods in the interpolation category do not explicitly allow for its incorporation into the merging process.

Integration-based methods generally provide a complete framework to incorporate both radar and rain gauge measurement uncertainties in the merging process, usually in the form of a covariance matrix. This has a direct impact on final merged estimates. For example, the BAY method explicitly accounts for the rain gauge's measurement and spatial sampling uncertainties, compares them with the radar's measurement uncertainty, and ultimately combines radar and rain gauge data in such a way that the overall uncertainty is minimized (Mazzetti & Todini, 2004; Todini, 2001). While the BAY method assumes that radar and rain gauge data are fully independent from each other, the CoK method enables accounting for the cross correlation between radar and rain gauge data and, as a result, between their measurement uncertainties. The potential benefits of accounting for the cross-correlation of radar and rain gauge error measurement uncertainties has been discussed by Ciach et al. (2003).

7.2. Preservation of Small-Scale (Convective) Rainfall Features

Since radar images are the main data source providing spatial details of the rainfall field, the ability of a given merging method to preserve small-scale rainfall features is highly related to the proportion of and way in which radar data are employed throughout the merging process. Other factors affecting the preservation of small-scale rainfall features and the effective correction of rainfall rates associated with them include whether the given features are captured by rain gauges (which in turn is a function of rain gauge network layout, density, and relative location for given storm conditions), along with key assumptions and mathematical constraints of the different merging methods.

As illustrated in Figure 3 (MFB), methods in the bias adjustment category essentially scale the original radar field so that rainfall accumulations match those recorded by rain gauges. As such, the original structure of the radar rainfall field is mostly preserved, including small-scale features. Considering this, bias adjustment methods can potentially provide a more complete representation of small-rainfall features as compared to methods from the other two categories. However, the fact that the bias adjustment is often applied uniformly across large areas (this is particularly the case for MFB, but even spatially variable bias adjustments include a degree of spatial averaging of radar-rain gauge biases), along with the fact that errors in radar QPEs are spatially variable and often rainfall rate-dependent, means that bias adjustment methods may fail to satisfactorily (quantitatively) correct the rainfall rates associated with these small-scale features (which, as discussed above, are often linked to local extremes).

Interpolation methods, on the other hand, are highly reliant upon rain gauge measurements, with spatial information from the radar field only considered at gauge locations and simply utilized to aid the interpolation process. In fact, merged outputs from interpolation-based methods often resemble the gauge-only interpolated rainfall field (see Figure 3). As a result and as discussed in previous sections, the final performance of interpolation-based merging, including the ability to reproduce small-scale features, is highly dependent upon the density of rain gauges and upon how well rain gauge records characterize the rainfall field (e.g., on whether or not rain gauges capture relevant rainfall features). Furthermore, as discussed in section 6.3, the Gaussian assumptions associated with geostatistical methods (including interpolation methods) often lead to smoothing of small-scale rainfall structures displaying high nonlinearities. This problem can be partially overcome through "normalization" of the radar field prior to merging (e.g., through data transformation or decomposition), followed by data reconstruction after merging. In particular, the singularity-based data decomposition (Wang, Ochoa-Rodríguez, Onof, & Willems, 2015) has shown great potential for improving preservation and reproduction of small-scale rainfall features throughout the merging process.

Lastly, integration methods undertake a true combination of the two data sources on the basis of their relative uncertainty, with the proportion of radar data utilized for merging generally falling in between that used in bias adjustment and interpolation methods. As a result of their uncertainty-based data combination, integration methods tend to take more information from the radar in areas where rain gauge SSE is larger (e.g., areas with limited rain gauge coverage and/or high spatial variability); this is visible from Figure 3. This is certainly a desirable feature (as long as radar QPEs are of a reasonable quality), which has the potential to lead to a better reproduction of small-scale features as compared to interpolation methods. However, same as interpolation methods, integration methods fall within the geostatistical category and are subject to Gaussian assumptions, which can result in smoothing of small-scale (non-Gaussian) rainfall structures. As discussed above, this shortcoming can be partially overcome through radar data decomposition.

7.3. Impact of Limited Availability of Rain Gauges on Merging Performance

As highlighted in section 5.2, the number and density of rain gauges employed in radar-rain gauge merging has a nonnegligible impact on merging performance. Having a limited number of rain gauges across the catchment area of interest may largely increase the chances of these not capturing relevant rainfall features (e.g., missing the core of the storm and/or missing small convective cells, the correct replication of which is critical for urban applications). This therefore increases the uncertainty associated with using rain gauge data as ground truth. More specifically, this increases the rain gauge SSE, with the impact being particularly pronounced at short aggregation times (Villarini, Mandapaka, et al., 2008; Wood et al., 2000a).

The impact of limited rain gauge availability on merging performance is closely linked to the reliance of a given merging method upon rain gauge data, as well as to the way in which radar data is employed in the merging process.

Considering their underlying theory and high reliance upon rain gauge records, interpolation-based methods are likely to be more sensitive to changes in rain gauge density in relation to other merging methods. This is indeed confirmed by available studies, such as that by Goudenhoofdt & Delobbe, 2009; see Figure 6b), where the performance of KED and KRE is shown to decrease quickly as the number of rain gauges decreases, while bias adjustment methods display a lower sensitivity to gauge density. It is worth noting that the study by Goudenhoofdt and Delobbe (2009) focuses on a large geographic domain (see Table 1). A preliminary urban-scale study by Bailey et al. (2014) further confirmed the higher sensitivity of KED to gauge density, in relation to MFB and BAY. Alternatives to partially overcome limitations in rain gauge data availability and therefore increased rain gauge SSE in interpolation-based merging methods include (1) use of pooled variograms, generated based on long-term rain gauge records (Schuurmans et al., 2007); (2) variogram generation based on the radar image rather than based on point rain gauge measurements (Erdin, 2013; Schuurmans et al., 2007); and (3) performing merging at coarser temporal resolution (e.g., 1 h) and then downscaling the merged results to the finer temporal resolution of interest (e.g., 5 min; Cecinati, de Niet, et al., 2017).

While generally displaying an inferior performance in relation to interpolation-based methods, bias adjustment methods are less sensitive to gauge density (see Figure 6b). The impact of limited rain gauge availability on bias adjustment method mainly derives from the increased uncertainty associated with rain gauge-based areal rainfall accumulations, which are used as unbiased rainfall estimates in the adjustment process. The impact of limited rain gauge availability on bias adjustment methods can however be partially offset by computing rain gauge-based areal rainfall accumulations (to be used as basis for adjustment) over prolonged periods, which, as discussed in section 2.1 and 6.5 can result in lower SSEs (Villarini, Mandapaka, et al., 2008; Wood et al., 2000a). It is possible to do so while still applying bias corrections at short-time intervals; for example, Harrison et al. (2009) apply MFB correction at 5-min intervals, but the radar-rain gauge bias adjustment factor is estimated based on hourly records.

Integration methods, on the other hand, have the ability to adapt to variations in rain gauge data availability. These methods undertake a true combination of radar and rain gauge data on the basis of their relative uncertainty. As a result, integration methods tend to take more information from the radar in areas where rain gauge SSE is larger (e.g., areas with limited rain gauge coverage and/or high spatial variability; see Figure 3). This is a desirable feature so long as radar quality is reasonable. Preliminary results by Bailey et al. (2014) suggest that the BAY integration method is less sensitive to rain gauge density than KED. In relation to MFB, the BAY method displays a similar sensitivity to rain gauge density (in terms of net change in performance measures with decreasing rain gauge density), while displaying an overall better performance. While integration methods are generally less sensitive to rain gauge data availability in comparison to interpolation methods, they are not immune to it. Similar strategies as those applicable to interpolation-based merging methods can be considered to partially offset the impact of limited rain gauge availability in the application of integration merging methods.

To the authors' knowledge, the only study looking at the impact of rain gauge data availability and density on merging performance at urban scales, and which includes all three types of merging methods, is that by Bailey et al. (2014). The results of this study are however limited in terms of storm events under consideration and level of analysis; further research is needed to better understand rain gauge density impact on merging performance at urban scales and, ideally, to identify critical rain gauge densities for urban scale applications, considering also variable storm conditions.

7.4. Computational Complexity and Requirements

In general, radar bias adjustment methods are of lower complexity than interpolation and integration methods; they are easier to implement, and their computational requirements are generally at the lower end of the spectrum. This is particularly the case for the simpler bias adjustment methods (e.g., MFB), the level of computational complexity of which is linear to the total number of radar pixels (m) and rain gauges (n) employed in the merging (i.e., computational complexity ~$O(n+m)$, where m is generally larger than n). The level of theoretical and computational complexity increases for the more sophisticated variations of radar bias adjustment methods, such as methods involving optimization and kriging interpolation of rain gauge records prior to merging, with the level of computational complexity of resolving the kriging interpolation system (of size $n \times n$) estimated to be ~$O(n^3)$ (Srinivasan et al., 2010).

Similarly, the computational "bottle-neck" of interpolation-based method lies in the solution of the kriging system, the computational complexity of which may increase slightly when including additional variables, besides the original rain gauge records. For example, the computational complexity of KED can be estimated to be ~$O((n+1)^3)$.

Integration methods are significantly more complex, both theoretically and computationally. Methods in this category require solving matrix systems, which are not only a function of n but also of m. In the case of the BAY method, kriging interpolation must be first performed on rain gauge records; afterward, the use of Kalman filter requires solving a matrix of size $m \times m$. As a result, the computational complexity of BAY can be estimated to be ~$O(n^3+m^3)$ (Montella, 2011). In the case of (full) CoK, a single matrix system of size $(n+m) \times (n+m)$ must be resolved; this results in a computational complexity of ~$O((n+m)^3)$. Considering that m is generally much larger than n, it is evident that the computational complexity of integration methods is significantly larger than that of bias adjustment and interpolation methods. What is more, the computational complexity of integration methods can be expected to increase significantly as a result of higher spatial resolution radar QPEs. In contrast, the computational complexity of bias correction and interpolation methods is mostly a function of n and therefore is not expected to increase largely with increasing spatial resolution of radar QPEs.

It is worth noting that the demanding computational requirements associated with the application of interpolation and integration methods can be partially overcome by splitting the merging domain into subdomains and performing merging at each of them (e.g., Mazzetti, 2004). This has two potential benefits: (1) reduction of the size of the matrices needing solving and (2) enabling parallelization of calculations. However, application of merging at subdomains poses new challenges, including selection of appropriate subdomain size and handling of merging at subdomain boundaries, so that the spatial continuity in the rainfall field is retained.

7.5. Summary Intercomparison of Merging Methods

Table 2 summarizes the main features and relative advantages and disadvantages of methods of each of the three application-oriented categories proposed in this review, with a focus on their potential for the urban hydrological applications of interest. The summary presented in this table is not only based upon the above discussion but also draws upon the main findings of the entire review.

8. Conclusions and Outlook

In this paper, an in-depth review of radar-rain gauge merging techniques was conducted, with a focus on their potential for the most stringent urban hydrological applications, such as sewer modeling, for which rainfall estimates of resolutions of the order of 1 km in space and 1-5 min in time are required. Drawing upon much of the literature on the topic of radar-rain gauge merging, the review explores the potential of merging at the fine scales of interest, considering specific challenges that may arise when applying radar-rain gauge merging at said scales. These include, among others, the fact that measurement errors and spatial-temporal sampling differences between radar and rain gauge measurements are magnified at fine scales, the need to preserve small-scale (e.g., convective) features in the merged estimates, the often limited density of rain gauges across urban areas, and the increasing computational requirements associated with the application of merging at higher spatial-temporal resolution.

The review starts with an overview and categorization of existing merging methods, following an application-oriented approach. The proposed merging categories are (1) methods focusing on bias adjustment of radar estimates (e.g., MFB adjustment and several variations of local bias adjustment); (2) rain gauge interpolation methods, which use radar spatial association as additional information (e.g., KED and kriging with radar-based error correction); and (3) radar and rain gauge integration methods (e.g., BAY and CoK). The main features of each type of merging method were described and discussed, and their relative performance was assessed based on inter-comparison of results reported in the literature (both at urban and larger scales). Likewise, key climatological, operational, and methodological factors impacting merging performance were identified and their relevance at the small spatial-temporal scales of interest was discussed.

Based on literature to date, it is clear that radar-rain gauge merging has the potential to significantly improve the quality and applicability of radar and rain gauge rainfall estimates for urban and other hydrological applications, despite literature on the application of merging at the urban scales of interest being limited. The general conclusion of studies to date is that radar-rain gauge merging always leads to superior rainfall estimates as compared to radar QPEs alone and—almost always, except in cases of very dense rain gauge networks (Biggs & Atkinson, 2011; Schuurmans et al., 2007)—as compared to rain gauge records alone.

Table 2
Intercomparison of the Three Categories of Merging Methods Identified in This Review

Feature	Merging category		
	Bias adjustment	Interpolation-based	Integration
General use of rain gauge and radar data	Rain gauge data are used as global or local unbiased rainfall estimates to correct radar rainfall accumulations; radar field is used as background and is therefore the main source of spatial rainfall structure.	Rain gauge data are assumed to be the best unbiased rainfall estimates[a] and are interpolated using spatial information from the radar image.	Rain gauge and radar rainfall estimates are proportionally combined at each location of interest, based on their relative uncertainty.
Relative performance (based on studies to date)	Generally outperformed by interpolation and integration methods	Generally better than bias adjustment methods; comparable to integration methods	Limited studies on testing of integration methods. Generally, performance is better than that of bias adjustment methods and comparable to that of interpolation methods.
Incorporation of measurement uncertainty in merging process[b]	Not accounted for in merging process	- Enable incorporation of rain gauge measurement uncertainty in the form of covariance - KRE explicitly accounts for rain gauge spatial sampling errors - Radar measurement error not explicitly accounted for	Enable incorporation of both rain gauge and radar measurement uncertainty
Preservation of small-scale rainfall features	Generally preserved, though corrections may be insufficient to correct high rain rates in convective areas.	Small-scale features may be smoothed out due to second-order statistical approximation.	Small-scale features may be smoothed out due to second-order statistical approximation.
Sensitivity to rain gauge density	Generally low	Generally high	Intermediate to low, with performance under low gauge density being highly dependent upon radar QPE quality
Sensitivity to radar QPE quality	Generally high	Relatively low	Intermediate
Model complexity	Generally low	Intermediate	High
Computational complexity[c]	Generally low $\sim 0(n+m)$	Intermediate $\sim 0((n+1)^3)$	High $\sim 0(n^3+m^3)$ $\sim 0((n+m)^3)$

Abbreviations: KRE, kriging with radar-based error correction; QPE, quantitative precipitation estimate.
[a]That is, minimum variance. [b]Leaving aside stochastic rainfall error models. [c]n: number of rain gauges employed in the merging process; m: number of radar pixels employed in the merging process.

The degree of improvement is a function of the merging method that is employed and of local climatological, infrastructural, and methodological factors. In fact, the potential of a given merging method for a given hydrological application is a function of multiple factors, including the climatological and operational (infrastructural) conditions in the catchment(s) of interest, the quality of the underlying radar and rain gauge records, the way in which both radar and rain gauge data are treated prior to merging, methodological choices related to merging application (e.g., variogram choice in the case of geostatistical methods, temporal aggregation scale, etc.), and the specific requirements of the application at hand (e.g., required spatial-temporal resolution of final rainfall estimates, whether merged rainfall estimates are required in real time or not, available computational resources, among others).

Each of the three merging categories under consideration have relative strengths and weaknesses, which may make them more or less suitable for a given application. On the whole, bias adjustment methods are at the lower end of the spectrum in terms of mathematical and computational complexity; this poses limitations in terms, for example, of handling of uncertainties in the underlying data sources, but has the advantage of easier implementation and less computational expense. As a result, methods in this category are the most widely used and tested. In contrast, integration methods are at the higher end of the spectrum, having the most complex mathematical formulation, allowing explicit incorporation of radar and rain gauge measurement uncertainties in the merging process and taking information from each source based on the associated certainty; however, these methods are the most computationally expensive, with computational requirements expected to grow exponentially with increasing radar QPE spatial resolution. The complexity and computational expense of integration methods makes them the least tested and adopted for operational uses. Interpolation-based methods represent a middle point in terms of complexity and computational requirements and are becoming increasingly popular, including for operational use (Jewell & Gaussiat, 2015). In terms of performance, bias adjustment methods generally display the poorest performance, being commonly outperformed by interpolation and integration methods. KED is the most popular and generally best performing interpolation method and has not only been shown to outperform bias adjustment methods but also the CoK integration method. The only method, which has displayed a similar or better performance than KED, is the BAY (integration) method. However, studies focusing on the testing and intercomparison of integration methods (including BAY and CoK) are very limited.

While the literature on radar-rain gauge merging is abundant and provides insights into the potential of merging at urban scales, studies to date have not sufficiently investigated important aspects related to radar-rain gauge merging applicability, particularly at the scales of interest. For example, only few studies include an intercomparison of various merging methods, with only two recent preliminary studies providing a comparison of the three categories of merging methods defined in the present paper (Kumar et al., 2016; Ochoa-Rodriguez, Wang, Bailey, et al., 2015). Furthermore, most of the studies focusing on intercomparison of merging methods and on analysis and handling of factors affecting merging performance have focused on large-scale applications, the resolution requirements of which are different from those of urban hydrological applications. While some of the studies focusing on methodological choices to improve merging performance have focused on urban scales (e.g., Cecinati, de Niet, et al., 2017; Wang, Ochoa-Rodríguez, Onof, & Willems, 2015; Wang, Ochoa-Rodríguez, van Assel, et al., 2015), these are limited to a single merging technique, often under limited climatological and infrastructural conditions.

Based on the present review and considering the identified research gaps, the following investigations are recommended in order to provide further evidence and comprehensive guidance on the applicability of merging methods, particularly at urban scales:

1. Methods from the three merging categories (i.e., bias adjustment, interpolation, and integration) should be intercompared, ideally in an urban context and under a range of storm conditions and infrastructural factors including gauge density and radar QPE quality. On grounds of performance and wide-spread use, it is recommended that the MFB method from the bias adjustment category, KED from the interpolation category, and BAY from the integration category be intercompared.
2. It is recommended that the three methodological choices identified as having the potential to improve merging performance at urban scales (i.e., temporal interpolation or advection correction of radar QPEs, radar data singularity-based decomposition prior to merging, and temporal aggregation of radar and rain gauge records prior to merging, followed by downscaling to the desired temporal resolution afterwards) be tested in combination with the three merging methods identified above (i.e., MFB, KED, and BAY) under a range of climatological and infrastructural conditions. It is worth mentioning that the singularity-based data decomposition is only suitable for methods involving kriging interpolation (e.g., KED and BAY) and cannot be applied in combination with MFB.

Looking into the future and considering advances in radar and other rainfall sensing technologies, as well as the increasing length of historical radar and rain gauge records, it is also recommended that research be conducted in the following areas:

1. Merging application under improved quality radar QPEs: as mentioned in section 2, Doppler, dual-polarization radars are becoming increasingly available and these additional radar hardware functionalities have the potential to lead to radar QPEs of increased accuracy (Berne & Krajewski, 2013; Bringi & Chandrasekar, 2001; Huuskonen et al., 2014; Wang et al., 2016). As discussed in this review, particularly in section 7, the quality of the underlying radar QPEs plays an important role in the quality of final merged estimates and on the relative performance of a given merging technique, depending on how information from radar QPEs is employed. In general, bias adjustment and integration methods utilize more information from radar QPEs than interpolation methods; as a result, they may be more sensitive to radar QPE quality and may therefore benefit more from improvements to it. While interpolation methods' use of radar QPE information is relatively smaller, they may in any case benefit from improvements to radar QPEs, particularly as these would improve the consistency between rain gauge and radar rainfall measurements, which, as widely discussed in section 6, is the most important single factor affecting the

performance of radar-rain gauge merging methods (Erdin, 2013). Ultimately, it is important to keep in mind that the quality of radar QPEs is likely to improve in the near future and it is critical to understand how this may impact the performance of different merging methods (bearing in mind that merging studies to date have generally used radar QPEs of inferior quality in comparison to what may be available in the near future).

2. Multisensor and multiscale rainfall data merging: nowadays, rainfall data from a variety of sources are becoming increasingly available, which have the potential to supplement rainfall measurements particularly across urban areas. These include, for example, data from local X-band radars, which supplement national radar networks; data from personal weather stations; and other novel ground rainfall sensors, including those deployed as part of citizen observatories to supplement ground rain gauge measurements; data from microwave links (Chandrasekar et al., 2009; de Vos et al., 2017; de Vos et al., 2018; Fencl et al., 2017; Hut, 2013; Muller et al., 2015; Pastorek et al., 2017; ten Veldhuis et al., 2014). Utilizing data from these alternative data sources poses new challenges, including the following:

a Data quality and how it is accounted for in the merging process. For example, data from citizen observatories are often of poor quality and, as such, may not be used directly as ground truth, but could in any case be used to supplement rainfall measurements across urban areas, provided that the uncertainty associated with it is adequately characterized and accounted for in the merging process.
b The need to combine more than two data sources, often at different spatial-temporal resolutions.
c The increasing computational requirements associated with merging more data sources, often at fine spatial-temporal resolutions.

Research is required to understand the potential of these data sources to supplement traditional radar and rain gauge networks and to explore ways of optimally combining multiple data source to obtain improved rainfall estimates across urban areas.

3. Data-driven radar-rain gauge merging approaches: it has already been several decades since radar QPEs started to be used operationally for meteorological and hydrological applications (Berne & Krajewski, 2013; Krajewski & Smith, 2002; Wilson & Brandes, 1979). As a result, radar QPE and associated rain gauge records are potentially long enough to enable development of data-driven radar-rain gauge merging methods. Interesting advances in this area are already reported in the literature (e.g., Chandrasekar et al., 2014; Kuang et al., 2016; Tang et al., 2018). This is certainly an area where more research is needed.
4. Multisource rainfall error model: rainfall data merging from radar and rain gauges and/or alternative data sources have the potential to improve final rainfall estimates. However, rainfall estimates will never be perfect and the true rainfall field is, by definition, unknown (Ciach & Krajewski, 1999a). Quantifying the residual errors in rainfall estimates is crucial in order to understand their reliability, as well as the impact that their uncertainty may have in subsequent runoff estimates. The quantification of errors in rainfall estimates has been an active topic of research for decades. However, existing rainfall error models have several shortcomings, including the fact that they are limited to describing errors associated with a single data source (e.g., errors associated with rain gauge measurements or radar QPEs alone or to rain gauge records alone) and to a single representative error source (e.g., radar-rain gauge differences and spatial temporal resolution; Ciach, 2003; Ciach & Krajewski, 1999a; Krajewski et al., 2010; Villarini & Krajewski, 2010; Villarini, Mandapaka, et al., 2008). New rainfall error models are therefore required, which (1) have the ability to incorporate multiple data sources and (2) enable explicit representation of different error sources (arising both from insufficient accuracy and resolution of rainfall estimates). Merging methods could potentially be used as starting point for error model development. In particular, the BAY merging method has the potential to fulfill the above-mentioned requirements, as it allows the explicit representation of uncertainties in the different data sources and combines them in such a way that the overall uncertainty is minimized.

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